Discrete Mathematics

CSE230

Discrete Mathematics and Its Applications

Kenneth H Rosen, New York: McGraw-Hill, 2019 (2003)

The book was sold more than 450K copies in US and is taught in more than 600 North American schools. It has been translated into Spanish, French, Portuguese, Greek, Chinese, Vietnamese and Korean.

This text is designed for a one- or two-term introductory discrete mathematics course taken by students in a wide variety of majors, including mathematics, computer science, and engineering. College algebra is the only explicit prerequisite, although a certain degree of mathematical maturity is needed to study discrete mathematics in a meaningful way.

Goals of a DM course

1. Mathematical Reasoning: to understand and construct mathematical reasoning starting with mathematical logic, mathematical induction
2. Combinatorial Analysis: ability of enumerating objects, counting
3. Discrete Structures: abstract mathematical structures like sets, permutations, relations, graphs, trees, finite state machines
4. Algorithmic Thinking: Some problems are solved by the specification of an algorithm, verification that it works properly and analysis of resource requirements
5. DM has applications to computer science, data networking, chemistry, biology, linguistics, geography, business and internet

CSE230 Topic Coverage

The course is designed with a focus on covering the first half of **Kenneth H. Rosen's textbook**. Studying the assigned textbook sections each week is **mandatory** for a student. Students should also utilize **practice problem-sets** to prepare for exams.

Specifically, this course will cover:

- ✓ Propositional Logic Basics (Chapter 1),
- ✓ Argumentative and Inductive Proofs (Chapter 1 and 5),
- ✓ Set Theory and Applications (Chapter 2),
- ✓ Series and Recurrence (Chapter 2 and 8),
- ✓ Number Theory and Applications (Chapter 4) and
- ✓ Basics of counting and combinatorics (Chapter 6).

Quiz Policy

There will be at least 5 quizzes over the duration of this course. Unless any unavoidable situation arises, there will be **one quiz every 3 classes**, taking some small time out of the lectures/class time. The final grade will be based on the **best N–2** quiz scores.

Reason for such frequent quizzes is the absence of assignments in the course. Regular quizzes are there to encouraged regular study habits in students. With the exception of extreme cases, there will be **no make-up quizzes for absentees**.

Assignment Policy

There will be at least 3 assignments over the duration of this course. The assignments are expected to be published/announced with a **deadline of one week**, requiring **handwritten solutions** of questions prepared by the faculty.

For scoring, all the assignments will be counted. Deadlines are to be treated strictly. However, a small extension may be allowed by the faculty, provided that the student had valid reasons and requested an extension before the deadline.

Course Resources

All resources:	
• Course Content Folder (Student):	http://tiny.cc/cse230fall25
• Official Discord Server:	https://discord.gg/[][][][] (Follow instructions from the bot after joining)
• Book Link:	https://drive.google.com/file/d/1apBrWJrTiOLrPpY-z1PxW8QQ-i5Pp6wI/view?usp=drive_link

Marks Distribution

Attendance	90%+ attendance = 5 marks	**5**
Assignments	Average of 3	**10**
Class work / Quizzes	Best N – 2, out of 5 (Minimum)	**20**
Midterm Exam	Converted from 50	**30**
Final Exam	Converted from 50	**35**
Bonus	3+3.5 bonus from midterm+final (capped)	**6.5**
TOTAL		**100+6.5**

There are some topics that are reserved as "self-study" topics. In both midterm and final exams, there will be **bonus questions** worth 5 marks that can be added to the respective midterm and final scores, without exceeding the total marks.

Lecture 1: Propositional Logic

Section 1.1

Propositions

- A *proposition* is a declarative sentence that is either true or false.
- Examples of propositions:
  a) The Moon is made of green cheese.
  b) Trenton is the capital of New Jersey.
  c) Toronto is the capital of Canada.
  d) $1 + 0 = 1$
  e) $0 + 0 = 2$
- Examples that are not propositions.
  a) Sit down!
  b) What time is it?
  c) $x + 1 = 2$
  d) $x + y = z$

Propositional Logic

- Propositions that cannot be expressed in terms of simpler propositions are called **atomic propositions**.

- Constructing Propositions
  - Propositional Variables: *p*, *q*, *r*, *s*, …
  - The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.

Propositional Logic

- Constructing Propositions
  - Propositional Variables: *p*, *q*, *r*, *s*, …
  - The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.
  - Compound Propositions: constructed from logical connectives and other propositions
    - Negation $\neg$
    - Conjunction $\wedge$
    - Disjunction $\vee$
    - Implication $\rightarrow$
    - Biconditional $\leftrightarrow$

Compound Propositions: Negation

Definition 1 Let p be a proposition. The *negation of p*, denoted by $\neg p$ (also denoted by $\overline{p}$), is the statement

"It is not the case that p."

The proposition $\neg p$ is read "not p." The truth value of the negation of p, $\neg p$, is the opposite of the truth value of p.

Remark: The notation for the negation operator is not standardized. Although $\neg p$ and $\overline{p}$ are the most common notations used in mathematics to express the negation of p, other notations you might see are $\sim p$, $-p$, p', Np, and $!p$.

Compound Propositions: Negation

- The *negation* of a proposition p is denoted by $\neg p$ and has this truth table:

p	$\neg p$
T	F
F	T

- **Example**: If p denotes "The earth is round.", then $\neg p$ denotes "It is not the case that the earth is round," or more simply "The earth is not round."

Conjunction & Disjunction

Definition 2

Let p and q be propositions. The *conjunction* of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Definition 3

Let p and q be propositions. The *disjunction* of p and q, denoted by $p \vee q$, is the proposition "p or q." The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Conjunction

- The *conjunction* of propositions p and q is denoted by $p \land q$ and has this truth table:

p	q	$p \land q$
T	T	T
T	F	F
F	T	F
F	F	F

- **Example**: If p denotes "I am at home." and q denotes "It is raining." then $p \land q$ denotes "I am at home and it is raining."

Disjunction

- The *disjunction* of propositions p and q is denoted by $p \vee q$ and has this truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- **Example**: If p denotes "I am at home." and q denotes "It is raining." then $p \vee q$ denotes "I am at home or it is raining."

The Connective Or in English

- In English "or" has two distinct meanings.
  - **"Inclusive Or"** - In the sentence "Students who have taken CSE230 or MAT120 may take this class," we assume that students need to have taken one of the prerequisites, but may have taken both. This is the meaning of disjunction. For $p \vee q$ to be true, either one or both of p and q must be true.
  - **"Exclusive Or"** - When reading the sentence "Soup or salad comes with this entrée," we do not expect to be able to get **both** soup and salad. This is the meaning of Exclusive Or (Xor). In $p \oplus q$, one of p and q must be true, but not both. The truth table for $\oplus$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Lecture 2: Propositional Logic

Section 1.1 (Continued)

Implication

Implication

- If p and q are propositions, then $p \rightarrow q$ is a *conditional statement* or *implication* which is read as "if p, then q" and has this truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- **Example**: If p denotes "I am at home." and q denotes "It is raining." then $p \rightarrow q$ denotes "If I am at home then it is raining."
- In $p \rightarrow q$, p is the *hypothesis* (*antecedent* or *premise*) and q is the *conclusion* (or *consequence*).

An example

"If you try hard for your exam, then you will succeed".

p = you tried hard for your exam.

q = you succeed

Case 1: "You tried hard for your exam" and "you succeed"

p = True q = True

Compound proposition p -> q is True

Case 2: "You tried hard for your exam" but "you failed"

p = True q = False

Compound proposition p -> q is False.

An example

Case 3: "You haven't tried hard for your exam" and "You succeeded"

p = False q = True

Compound proposition p -> q is True. Why?
because you can make the compound proposition false only when you satisfy the first condition itself i.e. p. If that itself not satisfied then we cannot make compound proposition False. Not False means True.

Case 4: "You haven't tried hard for your exam" and "You failed"

p = False q = False

Compound proposition p -> q is True. (Same reason as above)

Homework Problem:

Determine whether each of these conditional statements is True or False.

1. If $1 + 1 = 3$, then dogs can fly.
2. If $1 + 1 = 2$, then dogs can fly.
3. If monkeys can fly, then $1 + 1 = 3$
4. If $1 + 1 = 2$, then $2 + 2 = 5$
5. If Delhi is the capital of India then Beijing is the capital of China.

Understanding Implication

- In $p \to q$ there does not need to be any connection between the antecedent or the consequent. The "meaning" of $p \to q$ depends only on the truth values of p and q.
- These implications are perfectly fine, but would not be used in ordinary English.
  - "If the moon is made of green cheese, then I have more money than Bill Gates. "
  - "If the moon is made of green cheese then I'm on welfare."
  - "If $1 + 1 = 3$, then you will buy combat boots."

Understanding Implication (cont)

- Another way to view the logical conditional is to think of an obligation or contract.
  - "If I am elected, then I will lower taxes."
  - "If you get 100% on the final, then you will get an A."
- If the politician is elected and does not lower taxes, then the voters can say that he or she has broken the campaign pledge. Something similar holds for the professor. This corresponds to the case where p is true and q is false.

Different Ways of Expressing $p \rightarrow q$

- if p, then q
- if p, q
- q unless $\neg p$
- q if p
- q whenever p
- q follows from p
- p implies q
- p only if q
- q when p
- p is sufficient for q
- q is necessary for p

- a necessary condition for p is q
- a sufficient condition for q is p

How "if p then q" and "p only if q" can be same?

Example: "I will stay at home only if I'm sick."

Let p = "I will stay at home" and let q = "I'm sick"

Above statement is of the form p only if q

According to the above statement, becoming sick is the necessary condition that will make you stay at home.

This means "if you're not sick then, you cannot stay at home at any cost."

In order to falsify the above statement, q must be FALSE and p must be TRUE i.e. you are not sick and you still stay at home.

if p then q: "If I'll stay at home then I'm sick"

The only way to falsify the above statement is by making p TRUE and q FALSE. Therefore, p only if q is equivalent to if p then q

Converse, Contrapositive and Inverse

Converse, Contrapositive, and Inverse

- From $p \to q$ we can form new conditional statements .
  - $q \to p$ is the **converse** of $p \to q$
  - $\neg q \to \neg p$ is the **contrapositive** of $p \to q$
  - $\neg p \to \neg q$ is the **inverse** of $p \to q$

Example: Find the converse, inverse, and contrapositive of "Raining is a sufficient condition for me not going to town."

Solution:

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

Biconditional

- If p and q are propositions, then we can form the *biconditional* proposition $p \leftrightarrow q$, read as "p if and only if q." The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- If p denotes "I am at home." and q denotes "It is raining." then $p \leftrightarrow q$ denotes "I am at home if and only if it is raining."

Expressing the Biconditional

- Some alternative ways "p if and only if q" is expressed in English:
  - **p is necessary and sufficient for q**
  - **if p then q, and conversely**
  - **p iff q**

Truth Tables For Compound Propositions

- Construction of a truth table:
- Rows
  - Need a row for every possible combination of values for the atomic propositions.
- Columns
  - Need a column for the compound proposition (usually at far right)
  - Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
    - This includes the atomic propositions

Example Truth Table

- Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	¬r	p ∨ q	p ∨ q → ¬r
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent Propositions

- Two propositions are *equivalent* if they always have the same truth value.
- **Example**: Show using a truth table that the conditional is equivalent to the contrapositive.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Using a Truth Table to Show Non-Equivalence

Example: Show using truth tables that neither the converse nor inverse of an implication are not equivalent to the implication.

Solution:

p	q	$\neg p$	$\neg q$	$p \to q$	$\neg p \to \neg q$	$q \to p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

Problem

- How many rows are there in a truth table with *n* propositional variables?

  Solution: 2^n

- Note that this means that with n propositional variables, we can construct $2^{\wedge}2^n$ distinct (i.e., not equivalent) propositions.

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$ $\vee$	2 3
$\rightarrow$ $\leftrightarrow$	4 5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$
If the intended meaning is $p \vee (q \rightarrow \neg r)$
then parentheses must be used.

Lecture 3: Propositional Equivalence

Section 1.3

Tautologies, Contradictions, and Contingencies

- A *tautology* is a proposition which is always true.
  - Example: $p \vee \neg p$
- A *contradiction* is a proposition which is always false.
  - Example: $p \wedge \neg p$
- A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

P	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logical Equivalence: Examples

- Two compound propositions p and q are *logically equivalent* if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table shows that $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	**T**	**T**
T	F	F	**F**	**F**
F	T	T	**T**	**T**
F	F	T	**T**	**T**

Example: De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Augustus De Morgan

1806-1871

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	**F**	**F**
T	F	F	T	T	**F**	**F**
F	T	T	F	T	**F**	**F**
F	F	T	T	F	**T**	**T**

Example: Distributive Laws

$$(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$$

$$(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$$

This truth table shows that distributive law of disjunction over conjunction holds.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	**T**	T	T	**T**
T	T	F	F	**T**	T	T	**T**
T	F	T	F	**T**	T	T	**T**
T	F	F	F	**T**	T	T	**T**
F	T	T	T	**T**	T	T	**T**
F	T	F	F	**F**	T	F	**F**
F	F	T	F	**F**	F	T	**F**
F	F	F	F	**F**	F	F	**F**

Predicates and Quantifiers

Section 1.4

Section Summary

- Predicates
- Variables
- Quantifiers
  - Universal Quantifier
  - Existential Quantifier
- Negating Quantifiers
  - De Morgan's Laws for Quantifiers
- Translating English to Logic

Introducing Predicate Logic

- Predicate logic uses the following new features:
  - Variables: x, y, z
  - Predicates: $P(x)$, $M(x)$
  - Quantifiers (*to be covered in a few slides*):
- *Propositional functions* are a generalization of propositions.
  - They contain variables and a predicate, e.g., $P(x)$
  - Variables can be replaced by elements from their ***domain***.

Propositional Functions

- Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the *domain* (or *bound* by a quantifier, as we will see later).
- For example, let $P(x)$ denote "$x > 0$" and the domain be the integers. Then:
  - P(-3) is false.
  - P(0) is false.
  - P(3) is true.
- Often the domain is denoted by U. So in this example U is the integers.

Examples of Propositional Functions

- Let "$x + y = z$" be denoted by $R(x, y, z)$ and U (for all three variables) be the integers.
  Find these truth values:
  R(2,-1,5) **Solution: F**
  R(3,4,7) **Solution: T**
  R(x, 3, z) **Solution: Not a Proposition**

- Now let "$x - y = z$" be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:
  Q(2,-1,3) **Solution: T**
  Q(3,4,7) **Solution: F**
  Q(x, 3, z) **Solution: Not a Proposition**

Compound Expressions

- If $P(x)$ denotes "$x > 0$," find these truth values:

  $P(3) \lor P(-1)$ **Solution**: T

  $P(3) \land P(-1)$ **Solution**: F

  $P(3) \rightarrow P(-1)$ **Solution**: F

  $P(3) \rightarrow \neg P(-1)$ **Solution**: T

- Expressions with variables are not propositions and therefore do not have truth values. For example,

  $P(3) \land P(y)$

  $P(x) \rightarrow P(y)$

- When used with quantifiers (to be introduced next), these expressions (propositional functions) become propositions.

Quantifiers

Charles Peirce (1839-1914)

- We need *quantifiers* to express the meaning of English words including *all* and *some*:
  - "All men are Mortal."
  - "Some cats do not have fur."
- The two most important quantifiers are:
  - *Universal Quantifier,* "For all," symbol: $\forall$
  - *Existential Quantifier*, "There exists," symbol: $\exists$
- We write as in $\forall x\, P(x)$ and $\exists x\, P(x)$.
- $\forall x\, P(x)$ means $P(x)$ is true for <u>every</u> x in the *domain*.
- $\exists x\, P(x)$ means $P(x)$ is true for <u>some</u> x in the *domain*.
- The quantifiers are said to bind the variable x in these expressions.

Universal Quantifier

- $\forall x\, P(x)$ is read as "For all x, P(x)" or "For every x, P(x)"

  Examples:
  - If $P(x)$ denotes "$x > 0$" and U is the integers, then $\forall x\, P(x)$ is false.
  - If $P(x)$ denotes "$x > 0$" and U is the positive integers, then $\forall x\, P(x)$ is true.
  - If $P(x)$ denotes "x is even" and U is the integers, then $\forall x\, P(x)$ is false.

Existential Quantifier

- $\exists x\, P(x)$ is read as "For some x, P(x)", or as "There is an x such that P(x)," or "For at least one x, P(x)."

Examples:

- If $P(x)$ denotes "$x > 0$" and U is the integers, then $\exists x\, P(x)$ is true. It is also true if U is the positive integers.
- If $P(x)$ denotes "$x < 0$" and U is the positive integers, then $\exists x\, P(x)$ is false.
- If $P(x)$ denotes "x is even" and U is the integers, then $\exists x\, P(x)$ is true.

Thinking about Quantifiers

- When the domain is finite, we can think of quantification as looping through the elements of the domain.
- To evaluate $\forall x\, P(x)$ loop through all x in the domain.
  - If at every step $P(x)$ is true, then $\forall x\, P(x)$ is true.
  - If at a step $P(x)$ is false, then $\forall x\, P(x)$ is false and the loop terminates.
- To evaluate $\exists x\, P(x)$ loop through all x in the domain.
  - If at some step, $P(x)$ is true, then $\exists x\, P(x)$ is true and the loop terminates.
  - If the loop ends without finding an x for which $P(x)$ is true, then $\exists x\, P(x)$ is false.
- Even if the domains are infinite, we can still think of the quantifiers this fashion, but the loops will not terminate in some cases.

Properties of Quantifiers

- The truth value of $\exists x\, P(x)$ and $\forall x\, P(x)$ depend on both the propositional function $P(x)$ and on the domain U.
- **Examples**:
  - If U is the positive integers and $P(x)$ is the statement "$x < 2$", then **$\exists x\, P(x)$ is true, but $\forall x\, P(x)$ is false**.
  - If U is the negative integers and $P(x)$ is the statement "$x < 2$", then **both $\exists x\, P(x)$ and $\forall x\, P(x)$ are true**.
  - If U consists of 3, 4, and 5, and $P(x)$ is the statement "$x > 2$", then **both $\exists x\, P(x)$ and $\forall x\, P(x)$ are true**.
  - But if $P(x)$ is the statement "$x < 2$", then **both $\exists x\, P(x)$ and $\forall x\, P(x)$ are false**.

Precedence of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.
- For example, $\forall x\, P(x) \lor Q(x)$ means $(\forall x\, P(x)) \lor Q(x)$
- $\forall x\, (P(x) \lor Q(x))$ means something different.
- Unfortunately, often people write $\forall x\, P(x) \lor Q(x)$ when they mean $\forall x\, \mathbf{(}P(x) \lor Q(x)\mathbf{)}$.

Translating from English to Logic

Example 1: Translate the following sentence into predicate logic: "Every **student** in this class has taken a course in **Java**."

Solution:

First decide on the domain U.

Solution 1: If U is all students in this class, define a propositional function J(x) denoting "x has taken a course in Java" and translate as $\forall x\, J(x)$.

Solution 2: But if U is all people, also define a propositional function S(x) denoting "x is a student in this class" and translate as $\forall x\, (S(x) \rightarrow J(x))$.

$\forall x\, (S(x) \wedge J(x))$ is not correct. What does it mean?

Translating from English to Logic

Example 2: Translate the following sentence into predicate logic: "Some **student** in this class has taken a course in **Java**."

Solution:

First decide on the domain U.

Solution 1: If U is all students in this class, translate as

$$\exists x\, J(x)$$

Solution 2: But if U is all people, then translate as

$$\exists x\, (S(x) \land J(x))$$

$\exists x\, (S(x) \rightarrow J(x))$ is not correct. What does it mean?

Equivalences in Predicate Logic

- Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value
  - for every predicate substituted into these statements and
  - for every domain of discourse used for the variables in the expressions.
- The notation $S \equiv T$ indicates that S and T are logically equivalent.
- **Example**: $\forall x \neg\neg S(x) \equiv \forall x\, S(x)$

Thinking about Quantifiers as Conjunctions and Disjunctions

- If the domain is finite, a universally quantified proposition is equivalent to a conjunction of propositions without quantifiers and an existentially quantified proposition is equivalent to a disjunction of propositions without quantifiers.
- If U consists of the integers 1,2, and 3:

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

- Even if the domains are infinite, you can still think of the quantifiers in this fashion, but the equivalent expressions without quantifiers will be infinitely long.

Negating Quantified Expressions

- Consider $\forall x\, J(x)$

  "Every student in your class has taken a course in Java."

  Here $J(x)$ is "x has taken a course in Java" and

  the domain is students in your class.

- Negating the original statement gives "*It is not the case that* every student in your class has taken Java." This implies that "There is a student in your class who has not taken Java."

  Symbolically $\neg\forall x\, J(x)$ and $\exists x\, \neg J(x)$ are equivalent

Negating Quantified Expressions (*continued*)

- Now Consider $\exists x\, J(x)$

  "There is a student in this class who has taken a course in Java."

  Where $J(x)$ is "x has taken a course in Java."

- Negating the original statement gives "It is not the case that there is a student in this class who has taken Java." This implies that "Every student in this class has not taken Java"

  Symbolically $\neg\exists x\, J(x)$ and $\forall x\, \neg J(x)$ are equivalent

De Morgan's Laws for Quantifiers

- The rules for negating quantifiers are:

TABLE 2 **De Morgan's Laws for Quantifiers.**

Negation	*Equivalent Statement*	*When Is Negation True?*	*When False?*
$\neg\exists x P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

- The reasoning in the table shows that:

$$\neg\forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg\exists x P(x) \equiv \forall x \neg P(x)$$

Translation from English to Logic

Examples:

1. "Some student in this class has visited Mexico."

   Solution: Let $M(x)$ denote "x has visited Mexico" and $S(x)$ denote "x is a student in this class," and U be all people.

   $$\exists x \; (S(x) \land M(x))$$

1. "Every student in this class has visited Canada or Mexico."

   Solution: Add $C(x)$ denoting "x has visited Canada."

   $$\forall x \; (S(x) \rightarrow (M(x) \lor C(x)))$$

Some Translating from English into Logical Expressions

- U = {fast, slow, turning}

  $F(x)$: x is fast

  $S(x)$: x is slow

  $T(x)$: x is turning

  Translate "Everything is fast."

  Solution: $\forall x\, F(x)$

Translation (cont)

- U = {fast, slow, turning}

  $F(x)$: x is fast

  $S(x)$: x is slow

  $T(x)$: x is turning

  "Nothing is slow."

Solution: $\neg\exists x\, S(x)$ What is this equivalent to?

Solution: $\forall x \neg S(x)$

Translation (cont)

- U = {fast, slow, turning}

  $F(x)$: x is fast

  $S(x)$: x is slow

  $T(x)$: x is turning

"All fast things are slow."

Solution: $\forall x\,(F(x) \rightarrow S(x))$

Equivalent to: For all things, fast implies slow.

Translation (cont)

- U = {fast, slow, turning}

  $F(x)$: x is fast

  $S(x)$: x is slow

  $T(x)$: x is turning

"Some fast things are turning."

Solution: $\exists x\ (F(x) \wedge T(x))$

Equivalent to: Some things are fast and turning

Translation (cont)

- U = {fast, slow, turning}

  $F(x)$: x is fast

  $S(x)$: x is slow

  $T(x)$: x is turning

  "No slow thing is turning."

Solution: $\neg\exists x\, (S(x) \land T(x))$ What is this equivalent to?

Solution: $\forall x\, (\neg S(x) \lor \neg T(x))$

Translation (cont)

- U = {fast, slow, turning}

  F(x): *x* is fast

  S(x): *x* is slow

  T(x): x is turning

"If anything fast is a slow then it is also a turning."

Solution: $\forall x\,((F(x) \land S(x)) \rightarrow T(x))$

Charles Lutwidge Dodgson
(AKA Lewis Caroll)
(1832-1898)

Lewis Carroll Example

- The first two are called *premises* and the third is called the *conclusion*.
  - "All lions are fierce."
  - "Some lions do not drink coffee."
  - "Some fierce creatures do not drink coffee."
- Here is one way to translate these statements to predicate logic. Let P(x), Q(x), and R(x) be the propositional functions "x is a lion," "x is fierce," and "x drinks coffee," respectively.
  - $\forall x\,(P(x) \rightarrow Q(x))$
  - $\exists x\,(P(x) \wedge \neg R(x))$
  - $\exists x\,(Q(x) \wedge \neg R(x))$
- Later we will see how to prove that the conclusion follows from the premises.

Lecture 4: Introduction to Proofs

Section 1.7

Section Summary

- Mathematical Proofs
- Forms of Theorems
- Direct Proofs
- Indirect Proofs
  - Proof by Contraposition
  - Proof by Contradiction

Proofs of Mathematical Statements

- A ***proof*** is a valid argument that establishes the truth of a statement.

- Proofs have many practical applications:
  - verification that computer programs are correct
  - establishing that operating systems are secure
  - enabling programs to make inferences in artificial intelligence
  - showing that system specifications are consistent

Definitions

- A ***theorem*** is a statement that can be shown to be true using:
  - definitions
  - other theorems
  - *axioms* (statements which are given as true)
  - rules of inference
- A ***lemma*** is a 'helping theorem' or a result which is needed to prove a theorem.
- A ***corollary*** is a result which follows directly from a theorem.
- Less important theorems are sometimes called ***propositions***.
- A ***conjecture*** is a statement that is being proposed to be true. Once a proof of a conjecture is found, it becomes a theorem. It may turn out to be false.

Forms of Theorems

- Many theorems assert that a property holds for all elements in a domain, such as the integers, the real numbers, or some of the discrete structures that we will study in this class.

- Often the universal quantifier (needed for a precise statement of a theorem) is omitted by standard mathematical convention.

  For example, the statement:

  "If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$"

  really means

  "For all positive real numbers x and y, if $x > y$, then $x^2 > y^2$."

Proving Theorems

- Many theorems have the form:

$$\forall x(P(x) \rightarrow Q(x))$$

- To prove them, we show that: where c is an arbitrary element of the domain, $P(c) \rightarrow Q(c)$
- By universal generalization the truth of the original formula follows.
- So, we must prove something of the form: $p \rightarrow q$

Proving Conditional Statements: $p \to q$

- *Trivial Proof*: If we know q is true, then
  $p \to q$ is true as well.

"If it is raining then 1=1."

- *Vacuous Proof*: If we know p is false then
  $p \to q$ is true as well.

"If I am both rich and poor then $2 + 2 = 5$."

[Even though these examples seem silly, both trivial and vacuous proofs are often used in mathematical induction, as we will see in Chapter 5)]

Even and Odd Integers

Definition: The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k, such that $n = 2k + 1$. Note that every integer is either even or odd and no integer is both even and odd.

We will need this basic fact about the integers in some of the example proofs to follow. We will learn more about the integers in Chapter 4.

Proving Conditional Statements: $p \rightarrow q$

- ***Direct Proof***: Assume that p is true. Use rules of inference, axioms, and logical equivalences to show that q must also be true.

Example: Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Solution: Assume that n is odd. Then $n = 2k + 1$ for an integer k. Squaring both sides of the equation, we get:

$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1$,

where $r = 2k^2 + 2k$, an integer.

We have proved that if n is an odd integer, then n^2 is an odd integer. ◀

Proving Conditional Statements: $p \rightarrow q$

Definition: The real number r is *rational* if there exist integers p and q such that $r = p/q$, where $q \neq 0$.

Example: Prove that the sum of two rational numbers is rational.

Solution*:* Assume r and s are two rational numbers. Then there must be integers p, q and also t, u such that

$$r = p/q, \quad s = t/u, \quad u \neq 0, \; q \neq 0$$

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu+qt}{qu} = \frac{v}{w}$$

where $v = pu + qt$
$w = qu \neq 0$

Thus the sum is rational. ◀

Proof by Contraposition

- ***Proof by Contraposition***: Assume $\neg q$ and show $\neg p$ is true also. This is sometimes called an *indirect proof* method. If we give a direct proof of $\neg q \rightarrow \neg p$ then we have a proof of $p \rightarrow$ q.

  Why does this work?

Example: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution*:* Assume n is even. So, $n = 2k$ for some integer k. Thus,

$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2j$ for $j = 3k + 1$

Therefore $3n + 2$ is even. Since we have shown $\neg q \rightarrow \neg p$, $p \rightarrow q$ must hold as well. If n is an integer and $3n + 2$ is odd (not even) , then n is odd (not even).

Proof by Contraposition

Example: Prove that for an integer n, if n^2 is odd, then n is odd.

Solution: Use proof by contraposition. Assume n is even (i.e., not odd). Therefore, there exists an integer k such that $n = 2k$. Hence,

$$n^2 = 4k^2 = 2\,(2k^2)$$

and n^2 is even(i.e., not odd).

We have shown that if n is an even integer, then n^2 is even. Therefore by contraposition, for an integer n, if n^2 is odd, then n is odd.

◀

Proof by Contradiction

- ***Proof by Contradiction*****:** (AKA *reductio ad absurdum*).

To prove p (the theorem), assume $\neg p$ and derive a contradiction (such as $p \wedge \neg p$).

Since we have shown that $\neg p \rightarrow \mathbf{F}$ is true, it follows that the contrapositive $\mathbf{T} \rightarrow p$ also holds.

Example: Prove that if you pick 22 days from the calendar, at least 4 must fall on the same day of the week.

Solution: Assume that no more than 3 of the 22 days fall on the same day of the week. Because there are 7 days of the week, we could only have picked 21 days. This contradicts the assumption that we have picked 22 days. ◀

Proof by Contradiction

- **Example-1**: Use a proof by contradiction to give a proof that $\sqrt{2}$ is irrational.

  Solution: Suppose $\sqrt{2}$ is rational. Then there exists integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors (see Chapter 4). Then

  $$2 = \frac{a^2}{b^2} \qquad 2b^2 = a^2$$

  Therefore a^2 must be even. If a^2 is even then a must be even (an exercise). Since a is even, $a = 2c$ for some integer c. Thus,

  $$2b^2 = 4c^2 \qquad b^2 = 2c^2$$

  Therefore b^2 is even. Again then b must be even as well.

  But then 2 must divide both a and b. This contradicts our assumption that a and b have no common factors. We have proved by contradiction that our initial assumption must be false and therefore $\sqrt{2}$ is irrational. ◀

Proof by Contradiction

- **Example-2**: Give a proof by contradiction of the statement: “If $3n + 2$ is odd, then n is odd.”

  Solution: Let p be “$3n + 2$ is odd” and q be “n is odd.” To construct a proof by contradiction, assume that both p and $\neg q$ are true. That is, assume that $3n + 2$ is odd and that n is not odd. Because n is not odd, we know that it is even. -> Because n is even, there is an integer k such that $n = 2k$. This implies that $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. -> Because $3n + 2$ is $2t$, where $t = 3k + 1$, $3n + 2$ is even. Note that the statement “$3n + 2$ is even” is equivalent to the statement $\neg p$, because an integer is even if and only if it is not odd. -> Because both p and $\neg p$ are true, we have a contradiction. This completes the proof by contradiction, proving that if $3n + 2$ is odd, then n is odd.

Theorems that are Biconditional Statements

- To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, **we show that $p \rightarrow q$ and $q \rightarrow p$ are both true**.

Example: Prove the theorem: "If n is an integer, then n is odd if and only if n^2 is odd."

Solution: We have already shown (previous slides) that both $p \rightarrow q$ and $q \rightarrow p$. Therefore we can conclude $p \leftrightarrow q$.

Sometimes *iff* is used as an abbreviation for "if and only if," as in

"If n is an integer, then n is odd iff n^2 is odd."

Biconditional Statements are useful for proving equivalence

- **Example**: Show that these statements about the integer x are equivalent: (i) $3x + 2$ is even, (ii) $x + 5$ is odd, (iii) x^2 is even. *[Ch 1.7, Exercise 33]*

Solution: Let, p = "$3x + 2$ is even", q = "$x + 5$ is odd", and r = "x^2 is even".

Assume, $3x + 2$ is even. Therefore, $3x$ is even. [even – even = even]

Since, 3 is odd, x must be even.

Then, $x + 5$ must be odd. [even + odd = odd] (proved $p \rightarrow q$)

Also, x can be written as $2k$, where k is an integer.

Then $x^2 = (2k)^2 = 4k^2$. Thus, x^2 is even. (proved $p \rightarrow r$)

Now, we must prove the converse, $r \rightarrow p$ and $q \rightarrow p$. Complete it yourself.

Then we will have, $p \leftrightarrow q$ and $p \leftrightarrow r$. Therefore, $p \equiv q$ and $p \equiv r$.

In other words, $p \equiv q \equiv r$.

What is wrong with this?

"Proof" that *1 = 2*

Step	**Reason**
1. $a = b$	Premise
2. $a^2 = a \times b$	Multiply both sides of (1) by a
3. $a^2 - b^2 = a \times b - b^2$	Subtract b^2 from both sides of (2)
4. $(a - b)(a + b) = b(a - b)$	Algebra on (3)
5. $a + b = b$	Divide both sides by $a - b$
6. $2b = b$	Replace a by b in (5) because $a = b$
7. $2 = 1$	Divide both sides of (6) by b

Solution: Step 5. a - b = 0 by the premise
and division by 0 is undefined.

Assumptions are important.

What should be the assumption when we try to prove these statements?

- "if x is an irrational number and $x > 0$, then $\sqrt{x}$ is also irrational."
  - for direct proof: Assume x cannot be written as p/q, where $q \neq 0$. (Difficult to prove the statement, as we cannot explicitly represent irrationality in algebra)
  - for contraposition: Assume $\sqrt{x}$ is rational, therefore $\sqrt{x} = p/q$, where $q \neq 0$. (easier to proof for x from $\sqrt{x}$)

Assumptions are important.

What should be the assumption when we try to prove these statements?

- "If x, y, and z are integers and all three are even, then x +y +z is even,"
  - for direct proof: Assume x, y and z are evens, therefore, x = 2a, y = 2b and z = 2c. (Easier to find x+y+z)
  - for contraposition: Assume x+y+z = 2k + 1. (Relatively Lengthy, have to separately consider x, y and z and their cases)
- "if m and n are integers and mn is even, then m is even or n is even."
  - for direct proof: Assume mn is even, therefore mn = 2k. (Relatively Difficult, have to consider all cases of m and n)
  - for contraposition: Assume m is odd and n is odd, therefore m = 2a+1, n = 2b+1. (Easier to find mn)

Looking Ahead

- If direct methods of proof do not work:
  - We may need a clever use of a proof by contraposition.
  - Or a proof by contradiction.
  - Or try to ***disprove*** by counterexamples.
  - In Chapter 5, we will see mathematical induction and related techniques.

Lecture 5: Mathematical Induction

Section 5.1

Section Summary

- Mathematical Induction
- Examples of Proof by Mathematical Induction
- Mistaken Proofs by Mathematical Induction
- Guidelines for Proofs by Mathematical Induction

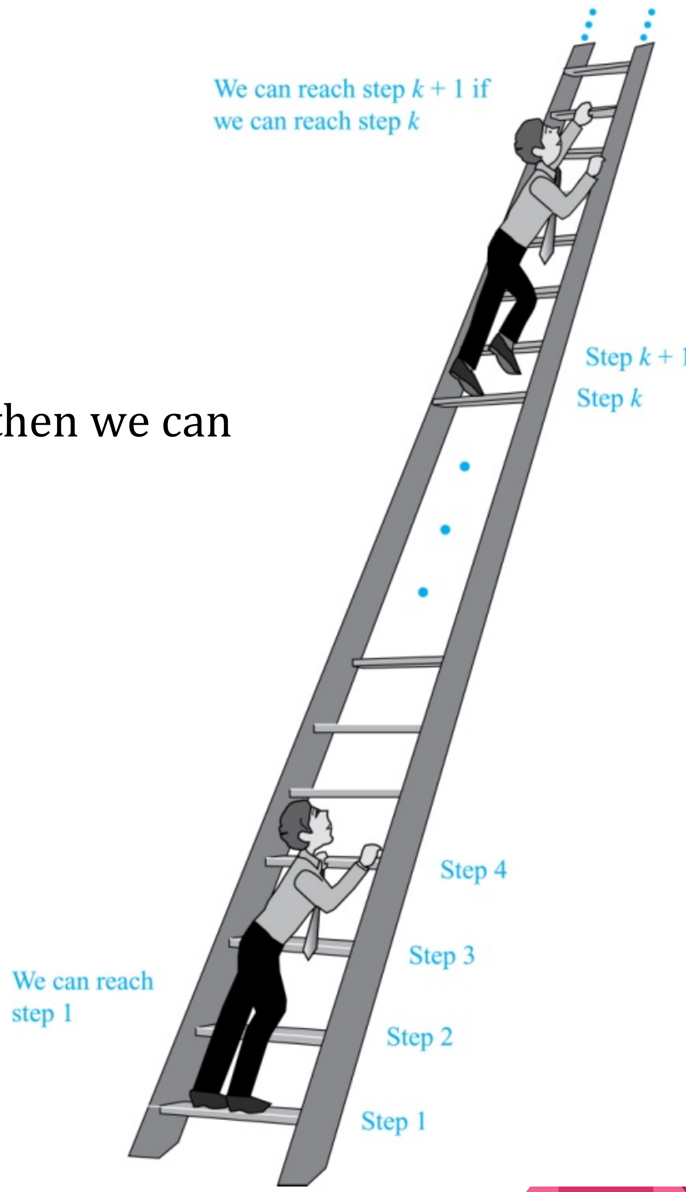
Climbing an Infinite Ladder

Suppose we have an infinite ladder:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

From (1), we can reach the first rung. Then by applying (2), we can reach the second rung. Applying (2) again, the third rung. And so on. We can apply (2) any number of times to reach any particular rung, no matter how high up.

This example motivates proof by mathematical induction.



Principle of Mathematical Induction

Principle of Mathematical Induction: To prove that $P(n)$ is true for all positive integers n, we complete these steps:

- *Basis Step*: Show that $P(1)$ is true.
- *Inductive Step*: Show that $P(k) \rightarrow P(k+1)$ is true for all positive integers k.

To complete the inductive step, assuming the *inductive hypothesis* that $P(k)$ holds for an arbitrary integer k, show that must $P(k+1)$ be true.

Climbing an Infinite Ladder Example:

- BASIS STEP: By (1), we can reach rung 1.
- INDUCTIVE STEP: Assume the inductive hypothesis that we can reach rung k. Then by (2), we can reach rung $k+1$.

Hence, $P(\mathrm{k}) \rightarrow P(\mathrm{k}+1)$ is true for all positive integers k. We can reach every rung on the ladder.

Important Points About Using Mathematical Induction

- Mathematical induction can be expressed as :

$$(P(1) \wedge \forall k\,(P(k) \rightarrow P(k+1))) \rightarrow \forall n\, P(n),$$

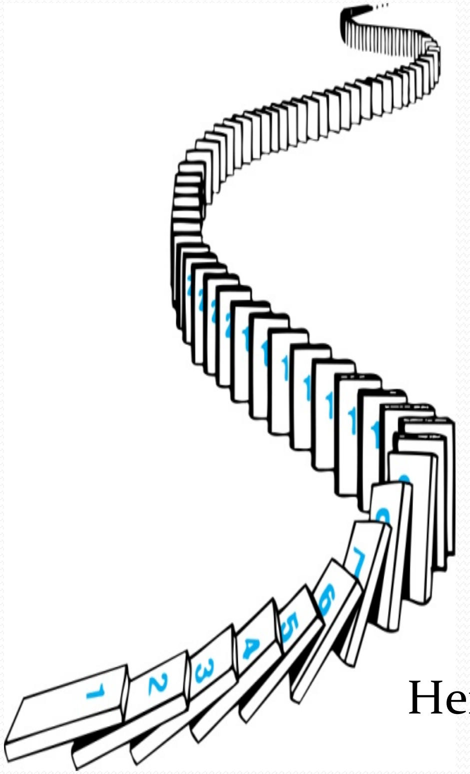
  where the domain is the set of positive integers.
- In a proof by mathematical induction, we **don't prove** that $P(k)$ is true for all positive integers! We show that <u>if we assume that $P(k)$ is true, then $P(k+1)$ must also be true</u>.

- Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer. We will see examples of this soon.

How Mathematical Induction Works

Consider an infinite sequence of dominoes, labeled 1,2,3, …, where each domino is standing.

Let $P(n)$ be the proposition that the nth domino is knocked over.



We know that the first domino is knocked down, i.e., $P(1)$ is true .

We also know that if whenever the k^{th} domino is knocked over, it knocks over the $(k + 1)^{\text{st}}$ domino, i.e, $P(k) \rightarrow P(k + 1)$ is true for all positive integers k.

Hence, all dominos are knocked over.

$P(n)$ is true for all positive integers n.

Proving a Summation Formula by Mathematical Induction

Example: Show that if n is a positive integer, then:

P(n): $$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Solution:

- BASIS STEP: $P(1)$ is true since $1 = 1(1 + 1)/2$.
- INDUCTIVE STEP: Assume $P(k)$ holds.

Then, the inductive hypothesis is: $$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

Under this assumption,

$$\begin{aligned} 1 + 2 + \ldots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$ ◀

Conjecturing and Proving Correct a Summation Formula

Example: Conjecture and prove correct a formula for the sum of the first n positive odd integers. Then prove your conjecture.

Solution: We have: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$,
$1 + 3 + 5 + 7 + 9 = 25$.

- We can conjecture that the sum of the first n positive odd integers is n^2,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2 .$$

- We prove that the conjecture is correct with mathematical induction.
- BASIS STEP: $P(1)$ is true since $1^2 = 1$.
- INDUCTIVE STEP: We have to show that, for every positive integer k,

$$P(k) \rightarrow P(k + 1).$$

Assume the inductive hypothesis $P(k)$ is True and then show that $P(k + 1)$ holds has well.

(continued →)

Conjecturing and Proving Correct a Summation Formula

Inductive Hypothesis: P(k) ≡ "$1 + 3 + 5 + \cdots + (2k - 1) = k^2$"

- So, assuming $P(k)$, it follows that:

$$
\begin{aligned}
&1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) \\
&= [1 + 3 + 5 + \cdots + (2k - 1)] + (2k + 1) \\
&= k^2 + (2k + 1) \quad [\textit{by the inductive hypothesis}] \\
&= k^2 + 2k + 1 = (k + 1)^2
\end{aligned}
$$

- Hence, we have shown that $P(k + 1)$ follows from $P(k)$. Therefore the sum of the first n positive odd integers is n^2.

◀

More Summation Proof by Induction

Example:

Show that: Sum of a finite number of terms in a geometric progression with initial term *a* and common ratio *r* is

$$\sum_{j=0}^{n} ar^j = a + ar + ar^2 + \cdots + ar^n = \frac{ar^{n+1} - a}{r - 1} \qquad \text{when } r \neq 1,$$

where n is a nonnegative integer.

Solution:

- BASIS STEP: $P(0)$ is true since

$$\frac{ar^{0+1} - a}{r - 1} = \frac{ar - a}{r - 1} = \frac{a(r - 1)}{r - 1} = a.$$

- INDUCTIVE STEP: Assun hypothesis would be:

$$a + ar + ar^2 + \cdots + ar^k = \frac{ar^{k+1} - a}{r - 1}.$$

Under such assumption, we add ar^{k+1} to both sides of the equality above.

More Summation Proof by Induction (contd.)

Rewriting the R.H.S of the equation gives

$$\frac{ar^{k+1} - a}{r - 1} + ar^{k+1} = \frac{ar^{k+1} - a}{r - 1} + \frac{ar^{k+2} - ar^{k+1}}{r - 1}$$

$$= \frac{ar^{k+2} - a}{r - 1}.$$

$$a + ar + ar^2 + \cdots + ar^k + ar^{k+1} = \frac{ar^{k+2} - a}{r - 1}.$$

which implies-

Conclusion: If the inductive hypothesis P(k) is true, then P(k+1) must be true. Proof by induction done!

Proving Divisibility Results

Example: Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n.

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

- BASIS STEP: $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.
- INDUCTIVE STEP: Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k. To show that $P(k+1)$ follows:

$$
\begin{aligned}
(k+1)^3 - (k+1) &= (k^3 + 3k^2 + 3k + 1) - (k+1) \\
&= (k^3 - k) + 3(k^2 + k)
\end{aligned}
$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. So by part (i) of Theorem 1 in Section 4.1 , $(k+1)^3 - (k+1)$ is divisible by 3.

Therefore, $n^3 - n$ is divisible by 3, for every integer positive integer n. ◀

Proving More Divisibility

Example: Use mathematical induction to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n.

Solution: Let P(n) denote the proposition: "$7^{n+2} + 8^{2n+1}$ is divisible by 57."

- BASIS STEP: $P(0)$ is true since $7^{0+2} + 8^{2\cdot 0+1} = 7^2 + 8^1 = 57$ is divisible by 57.
- INDUCTIVE STEP: Assume $P(k)$ holds, that is, we assume that $7^{k+2} + 8^{2k+1}$ is divisible by 57. So we can write: $7^{k+2} + 8^{2k+1} = 57x$, where x is an integer.

  Then, we must show that $P(k+1)$, the statement that "$7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57", is also true. We find that,

  $7^{(k+1)+2} + 8^{2(k+1)+1} = 7^{k+3} + 8^{2k+3} = 7 \cdot 7^{k+2} + 8^2 \cdot 8^{2k+1} = 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1}$

  $= 7(\mathbf{7^{k+2} + 8^{2k+1}}) + 57 \cdot 8^{2k+1} = 7(\mathbf{57x}) + 57 \cdot 8^{2k+1} = 57(7x + 8^{2k+1})$.

  Therefore, $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57. ◀

An Incorrect "Proof" by Mathematical Induction (Cont.)

Example: Let $P(n)$ be the statement that "Every set of n lines in the plane, no two of which are parallel, meet in a common point." Here is a "proof" that $P(n)$ is true for all positive integers $n \geq 2$.

- BASIS STEP: The statement $P(2)$ is true because any two lines in the plane that are not parallel meet in a common point.
- INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true for the positive integer $k \geq 2$, i.e., every set of k lines in the plane, no two of which are parallel, meet in a common point.

- We must show that if $P(k)$ holds, then $P(k + 1)$ holds,
- i.e., if every set of k lines in the plane, no two of which are parallel, $k \geq 2$, meet in a common point,
- then every set of k + 1 lines in the plane, no two of which are parallel, meet in a common point.

continued →

An Incorrect "Proof" by Mathematical Induction (Cont.)

Inductive Hypothesis: Every set of k lines in the plane, where $k \geq 2$, no two of which are parallel, meet in a common point.

- Consider a set of $k + 1$ distinct lines in the plane, no two parallel. By the inductive hypothesis, the first k of these lines must meet in a common point p_1. By the inductive hypothesis, the last k of these lines meet in a common point p_2.
- If p_1 and p_2 are different points, all lines containing both of them must be the same line since two points determine a line. This contradicts the assumption that the lines are distinct. Hence, $p_1 = p_2$ lies on all $k + 1$ distinct lines, and therefore $P(k + 1)$ holds. Assuming that $k \geq 2$, distinct lines meet in a common point, then every $k + 1$ lines meet in a common point.
- There must be an error in this proof since the conclusion is absurd. But where is the error?
  - **Answer**: $P(k) \rightarrow P(k + 1)$ only holds for $k \geq 3$. It is not the case that $P(2)$ implies $P(3)$. The first two lines must meet in a common point p_1 and the second two must meet in a common point p_2. They do not have to be the same point since only the second line is common to both sets of lines.

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form "for all $n \geq b$, $P(n)$" for a fixed integer b. For statements of the form "$P(n)$ for all positive integers n," let $b = 1$, and for statements of the form "$P(n)$ for all nonnegative integers n," let $b = 0$. For some statements of the form $P(n)$, such as inequalities, you may need to determine the appropriate value of b by checking the truth values of $P(n)$ for small values of n, as is done in Example 6.
2. Write out the words "Basis Step." Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words "Inductive Step" and state, and clearly identify, the inductive hypothesis, in the form "Assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$."
4. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
5. Prove the statement $P(k + 1)$ making use of the assumption $P(k)$. (Generally, this is the most difficult part of a mathematical induction proof. Decide on the most promising proof strategy and look ahead to see how to use the induction hypothesis to build your proof of the inductive step. Also, be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k, including $k = b$.)
6. Clearly identify the conclusion of the inductive step, such as by saying "This completes the inductive step."
7. After completing the basis step and the inductive step, state the conclusion, namely, "By mathematical induction, $P(n)$ is true for all integers n with $n \geq b$".

Lecture 6: Sets

Section 2.1

Section Summary

- Definition of sets
- Describing Sets
  - Roster Method
  - Set-Builder Notation
- Some Important Sets in Mathematics
- Empty Set and Universal Set
- Subsets and Set Equality
- Cardinality of Sets
- Tuples
- Cartesian Product

Sets

- A *set* is an unordered collection of objects.
  - the students in this class
  - the chairs in this room
- The objects in a set are called the *elements*, or *members* of the set. A set is said to *contain* its elements.
- The notation $a \in A$ denotes that a is an element of the set A.
- If a is not a member of A, write $a \notin A$

Describing a Set: Roster Method

- $S = \{a,b,c,d\}$
- Order not important

  $S = \{a,b,c,d\} = \{b,c,a,d\}$
- Each distinct object is either a member or not; listing more than once does not change the set.

  $S = \{a,b,c,d\} = \{a,b,c,b,c,d\}$
- Elipses (…) may be used to describe a set without listing all of the members when the pattern is clear.
- $S = \{a,b,c,d, \ldots\ldots,z\}$

Roster Method

- Set of all vowels in the English alphabet:

  $V = \{a,e,i,o,u\}$

- Set of all odd positive integers less than 10:

  $O = \{1,3,5,7,9\}$

- Set of all positive integers less than 100:

  $S = \{1,2,3,........,99\}$

- Set of all integers less than 0:

  $S = \{...., -3,-2,-1\}$

Some Important Sets

- $\mathbf{N} = \textit{natural numbers} = \{0,1,2,3 \ldots.\}$
- $\mathbf{Z} = \textit{integers} = \{\ldots,-3,-2,-1,0,1,2,3,\ldots\}$
- $\mathbf{Z}^{+} = \textit{positive integers} = \{1,2,3,\ldots..\}$
- $\mathbf{R}$ = set of *real numbers*
- $\mathbf{R}^{+}$ = set of *positive real numbers*
- $\mathbf{C}$ = set of *complex numbers*.
- $\mathbf{Q}$ = set of rational numbers

Set-Builder Notation

- Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^{+} \mid x \text{ is odd and } x < 10\}$$

- A predicate may be used:

$$S = \{x \mid \mathrm{P}(x)\}$$

- Example: $S = \{x \mid \mathrm{Prime}(x)\}$
- Positive rational numbers:

$$\mathbf{Q}^{+} = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p,q\}$$

Interval Notation

- $[a,b] = \{x \mid a \le x \le b\}$
- $[a,b) = \{x \mid a \le x < b\}$
- $(a,b] = \{x \mid a < x \le b\}$
- $(a,b) = \{x \mid a < x < b\}$

closed interval [a,b]

open interval (a,b)

Universal Set and Empty Set

- The *universal set U* is the set containing everything currently under consideration.
  - Sometimes implicit
  - Sometimes explicitly stated.
  - Contents depend on the context.
- The *empty set* is the set with no elements.
  - Symbolized as $\emptyset$ or, $\{\}$.

Venn Diagrams

- The universal set is represented by a rectangle.
- Sets are shown as ovals, circles or other geometric shapes.
- Sometimes points are used to represent the particular elements of the set.

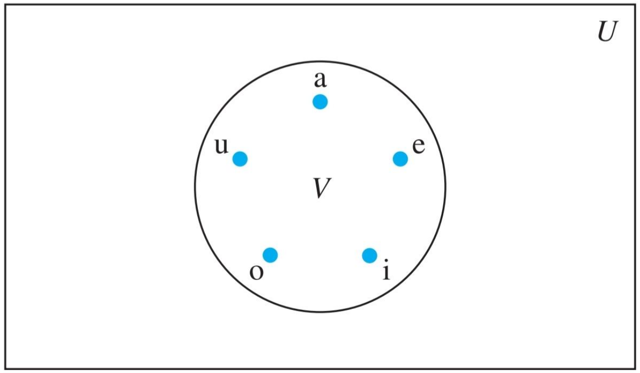
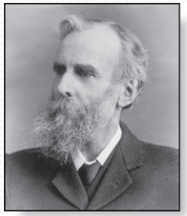


FIGURE 1 **Venn diagram for the set of vowels.**



John Venn (1834-1923)
Cambridge, UK

Some things to remember

- Sets can be elements of sets.

  $\{\{1,2,3\}, a, \{b,c\}\}$

  $\{\mathbf{N},\mathbf{Z},\mathbf{Q},\mathbf{R}\}$

- The empty set is different from a set containing the empty set.

  $\emptyset \neq \{ \emptyset \}$

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$.
- We write $A = B$ if A and B are equal sets.

$$\{1,3,5\} = \{3, 5, 1\}$$

$$\{1,5,5,5,3,3,1\} = \{1,3,5\}$$

Subsets

Definition: The set A is a *subset* of B, if and only if every element of A is also an element of B.

- The notation $A \subseteq B$ is used to indicate that A is a subset of the set B.
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.
  - $\emptyset \subseteq S$, for every set S (Because $a \in \emptyset$ is always false).
  - $S \subseteq S$, for every set S (Because $a \in S \rightarrow a \in S$).

Showing a Set is or is not a Subset of Another Set

- **Showing that A is a Subset of B**: To show that $A \subseteq B$, show that if x belongs to A, then x also belongs to B.
- **Showing that A is not a Subset of B**: To show that A is not a subset of B, $A \nsubseteq B$, find an element $x \in A$ with $x \notin B$. (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples:

- The set of all computer science majors at your school is a subset of all students at your school.
- The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

Another look at Equality of Sets

- Recall that two sets A and B are *equal*, denoted by $A = B$, iff
$$\forall x(x \in A \leftrightarrow x \in B)$$
- Using logical equivalences we have that $A = B$ iff
$$\forall x[(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$
- This is equivalent to
$$A \subseteq B \qquad \text{and} \qquad B \subseteq A$$

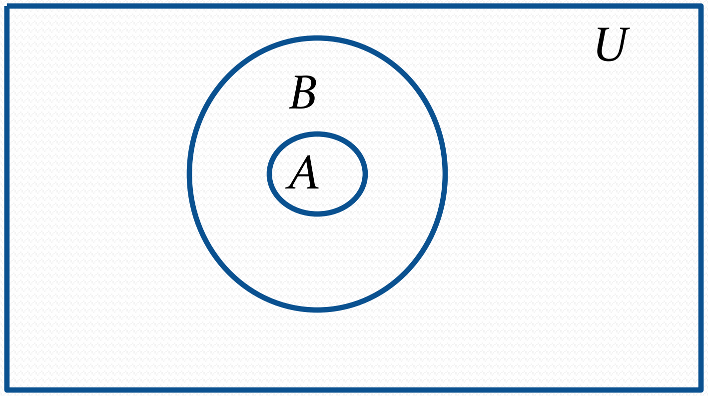
Proper Subsets

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B, denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \to x \in B) \land \exists x(x \in B \land x \notin A)$$

is true.

Venn Diagram:



Set Cardinality

If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* of a finite set A, denoted by $|A|$, is the number of (distinct) elements of A.

Examples:

1. $|ø| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{ø\}| = 1$
5. The set of integers is infinite.

Power Sets

Definition: The set of all subsets of a set A, denoted $P(A)$, is called the *power set* of A.

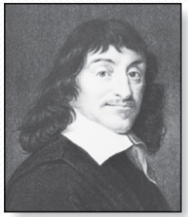
Example: If $A = \{a, b\}$ then

$$P(A) = \{ø, \{a\}, \{b\}, \{a, b\}\}$$

Example: If $A = \{ø, \{a, b\}\}$ then

$$P(A) = \{ø, \{ø\}, \{\{a, b\}\}, \{ø, \{a, b\}\}\}$$

- If a set has n elements, then the cardinality of the power set is 2^n.



René Descartes
(1596-1650)

Cartesian Product

Definition: The *Cartesian Product* of two sets A and B, denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Example:

$A = \{a,b\}$ $B = \{1,2,3\}$

$A \times B = \{(a,1),(a,2),(a,3), (b,1),(b,2),(b,3)\}$

- **Definition**: A subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B.

Cartesian Product

Definition: The cartesian products of the sets $A_1, A_2, \ldots\ldots, A_n$ denoted by $A_1 \times A_2 \times \ldots\ldots \times A_n$, is the set of ordered n-tuples $(a_1, a_2, \ldots\ldots, a_n)$ where a_i belongs to A_i for $i = 1, \ldots n$.

$$A_1 \times A_2 \times \cdots \times A_n = \\ \{(a_1, a_2, \ldots, a_n) | a_i \in A_i \text{ for } i = 1, 2, \ldots n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution: $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$

Lecture 7:
Set Operations

Section 2.2

Section Summary

- Set Operations
  - Union
  - Intersection
  - Complementation
  - Difference
- Set Cardinality: Inclusion, Exclusion
- Set Identities
- Proving Identities

Union

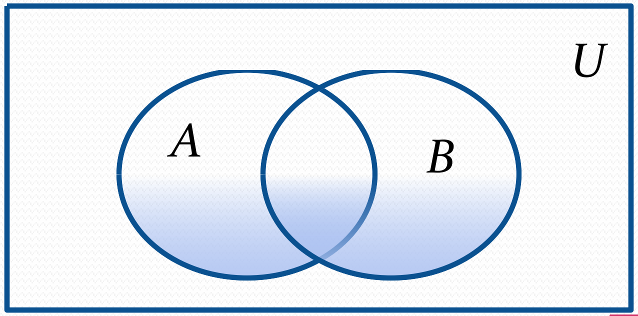
- **Definition**: Let A and B be sets. The *union* of the sets A and B, denoted by $A \cup B$, is the set:

$$\{x | x \in A \lor x \in B\}$$

- **Example**: What is $\{1,2,3\} \cup \{3, 4, 5\}$?

Solution: $\{1,2,3,4,5\}$

Venn Diagram for $A \cup B$



Intersection

- **Definition**: The *intersection* of sets A and B, denoted by $A \cap B$, is

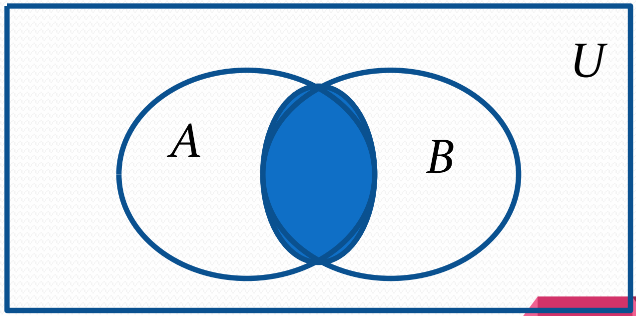
$$\{x | x \in A \wedge x \in B\}$$

- Note if the intersection is empty, then A and B are said to be *disjoint*.
- **Example**: What is $\{1,2,3\} \cap \{3,4,5\}$?

  Solution: $\{3\}$
- **Example:** What is $\{1,2,3\} \cap \{4,5,6\}$?

  Solution: $\emptyset$

Venn Diagram for $A \cap B$



Complement

Definition: If A is a set, then the complement of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$.

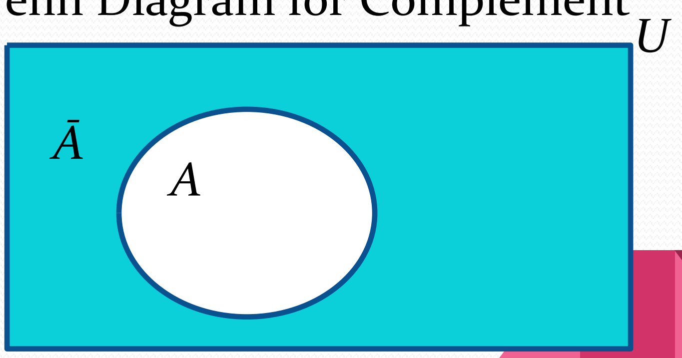
$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c.)

Example: If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$?

Solution: $\{x \mid x \leq 70\}$

Venn Diagram for Complement

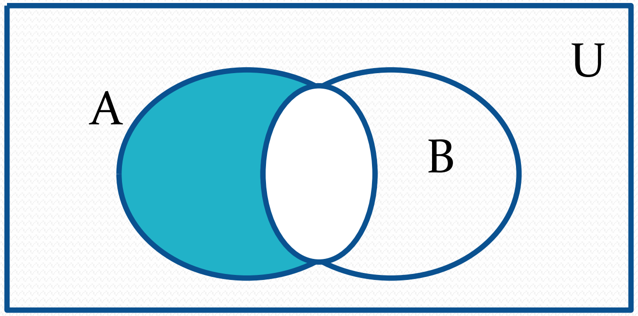


Difference

- **Definition**: Let *A* and *B* be sets. The *difference* of *A* and *B*, denoted by *A* – *B*, is the set containing the elements of *A* that are not in *B*. The difference of *A* and *B* is also called the complement of *B* with respect to *A*.

$$A - B = \{x \mid x \in A \land x \notin B\} = A \cap B^c$$

Venn Diagram for $A - B$



Review Questions

Example: $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

a. $A \cup B$

   Solution: $\{1,2,3,4,5,6,7,8\}$

a. $A \cap B$

   Solution: $\{4,5\}$

a. $\bar{A}$

   Solution: $\{0,6,7,8,9,10\}$

a. $\bar{B}$

   Solution: $\{0,1,2,3,9,10\}$

a. $A - B$

   Solution: $\{1,2,3\}$

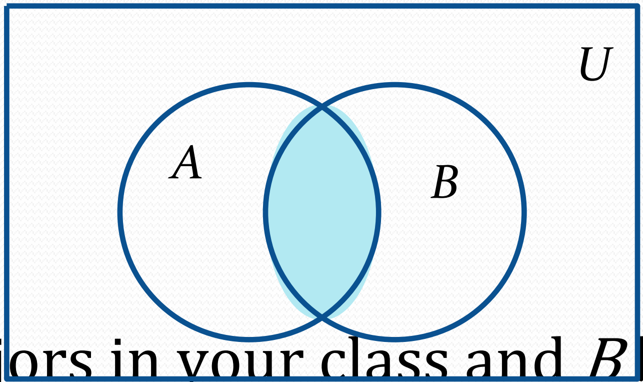
a. $B - A$

   Solution: $\{6,7,8\}$

The Cardinality of the Union of Two Sets

- Inclusion-Exclusion Principle

$$|A \cup B| = |A| + |B| - |A \cap B|$$



- **Example**: Let A be the math majors in your class and B be the CS majors. To count the number of students who are either math majors or CS majors:
  - Add the number of math majors and the number of CS majors.
  - Then subtract the number of joint CS/math majors.

Inclusion-Exclusion Principle

Example: A restaurant served 50 guests one day. At the end, there was 37 orders of burgers and 29 orders of smoothies. If everyone ordered a burger or a smoothie that day, how many people ordered both?

Solution:

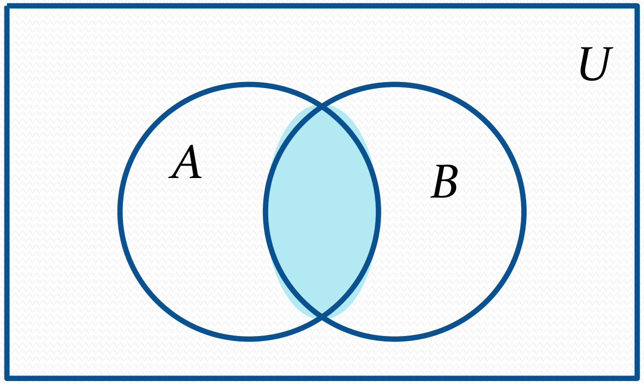
Let A and B be the sets of people who ordered burgers and smoothies respectively. Then, $|A| = 37$ and $|B| = 29$.

Since, $|A \cup B| = 50$

$=> \ |A| + |B| - |A \cap B| = 50$

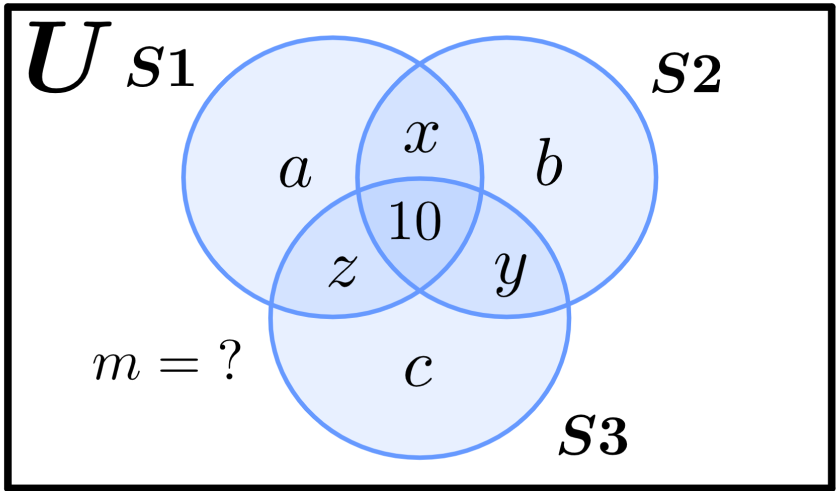
$=> |A \cap B| = |A| + |B| - 50$

$=> |A \cap B| = 37 + 29 - 50 = 16.$



Inclusion-Exclusion Principle

Example: A coding competition had 100 students participating. 50 of the participants were able to solve the first problem. However, for the second and third problem, 39 and 23 students were able to solve them, respectively. **If 10 students solved all three problems and 12 solved any 2, how many students solved none**?



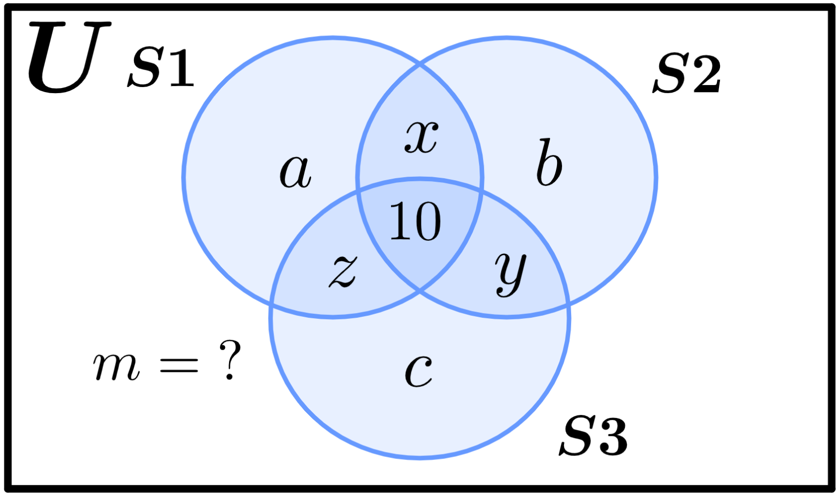
Solution: Let, S1, S2 and S3 be the sets of students who answered the first, second and third problem respectively, & U be the universal set of all participants.

Then, $|S1| = 50 = a+x+z+10$,
$|S2| = 39 = b+x+y+10$, $|S3| = 23 = c+y+z+10$ and $|U| = 100$.
Inferring from the Venn Diagram,

Number of students that solved zero problems = m,
Number of students that solved 1 problem = a+b+c,
Number of students that solved 2 problems = x+y+z = 12 (given),
Number of students that solved all 3 problems = 10.

Inclusion-Exclusion Principle

Example: A coding competition had 100 students participating. 50 of the participants were able to solve the first problem. However, for the second and third problem, 39 and 23 students were able to solve them, respectively. **If 10 students solved all three problems and 12 solved any 2, how many students solved none**?



Solution (continued):

Here, |S1|+|S2|+|S3| = 50+39+23

=> (a+x+z+10)+(b+x+y+10)+
(c+y+z+10) = 112

=> a+b+c+2x+2y+2z = 82

=> a+b+c = 82 – 2(x+y+z)

=> a+b+c = 82 – 2*12 = 58

So, 58 students solved exactly 1 problem.

Also, |U| = m+(a+b+c)+(x+y+z)+10 = 100.

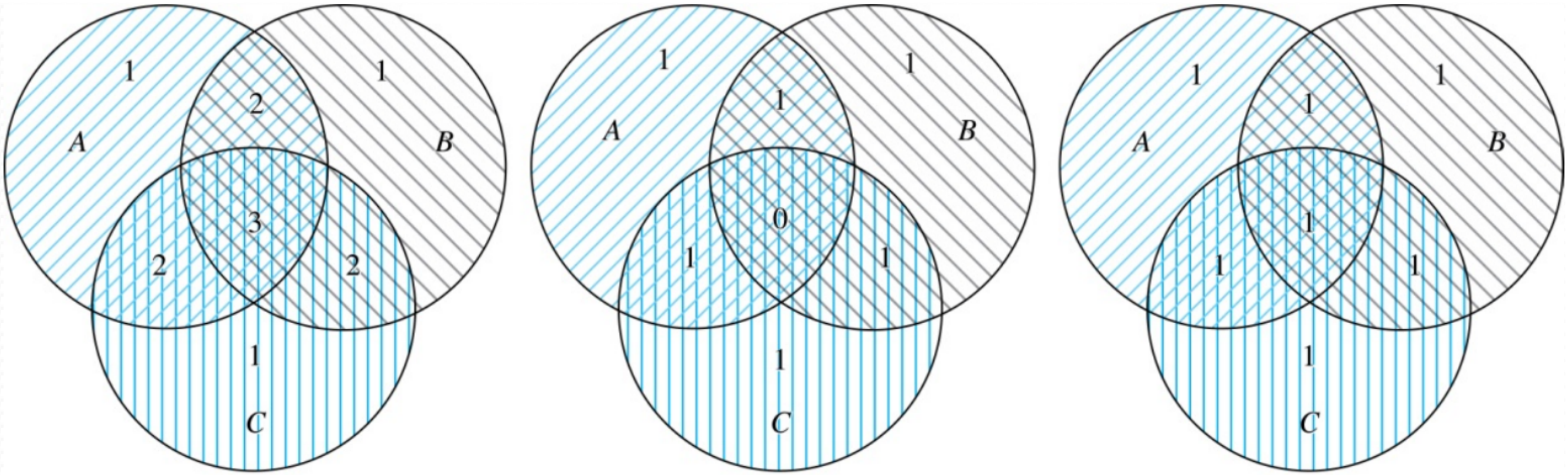
=> m+58+12+10=100 => m = 100–80 = 20.

Thus, 20 students solved zero problems.

Inclusion-Exclusion Principle for 3 Sets

- For 3 sets, the Inclusion-Exclusion Principle becomes:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$



(a) Count of elements by $|A|+|B|+|C|$

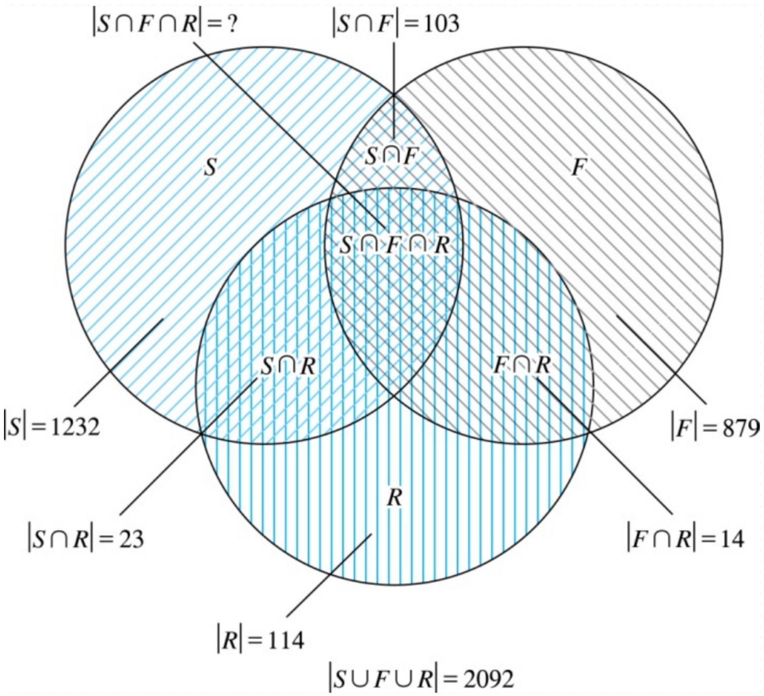
(b) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|$

(c) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$

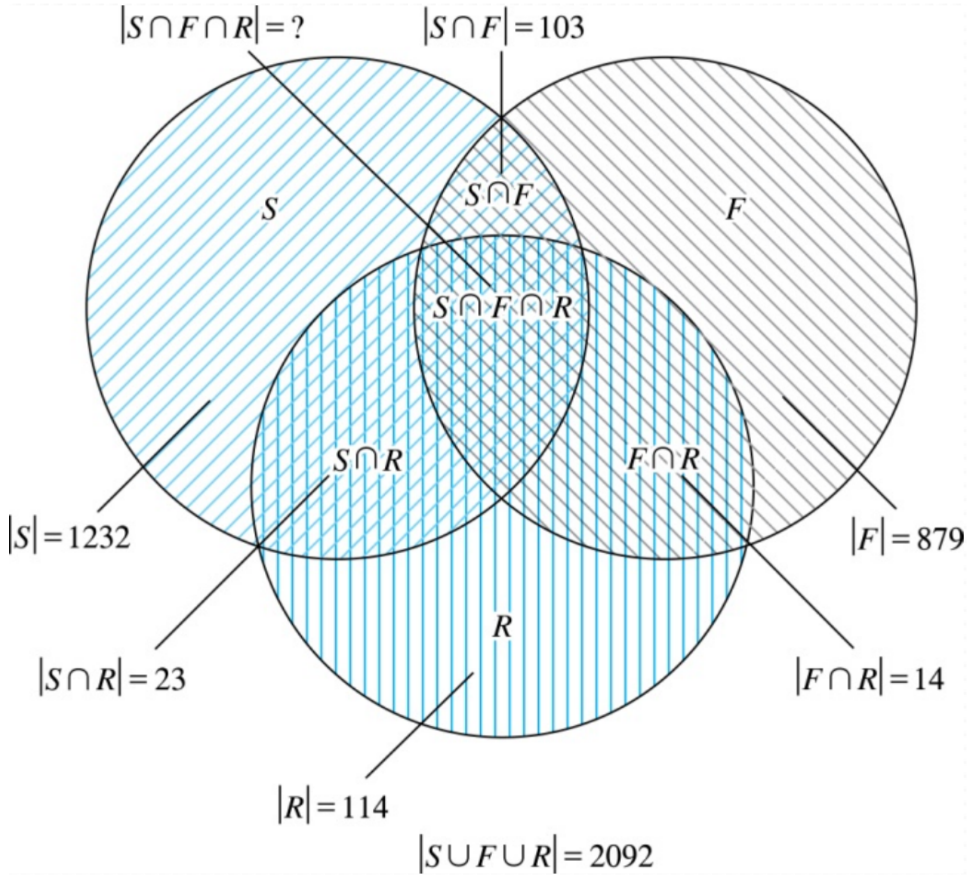
Inclusion-Exclusion Principle for 3 Sets

- **Example:** In a class,

1232 students have taken a course in Spanish,
879 have taken a course in French, and
114 have taken a course in Russian. Further,
103 have taken courses in both
Spanish and French,
23 have taken courses in both
Spanish and Russian, and
14 have taken courses in both
French and Russian.
If 2092 students have taken a course in
at least one of Spanish, French and Russian,
how many students have taken a course in all 3 languages?



Inclusion-Exclusion Principle for 3 Sets



- **Example (Cont.):**

Here, $|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|$

$\Rightarrow 2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|.$

$\Rightarrow |S \cap F \cap R| = 7.$

Set Identities

TABLE 1 **Set Identities.**	
Identity	***Name***
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Proving Set Identities

- Different ways to prove set identities:
  1. Prove that **each set (side of the identity) is a subset of the other**.
  2. Use set builder notation and **propositional logic**.
  3. **Membership Tables:** Verify that elements in the same combination of sets always either belong or do not belong to the same side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not.

Proving Set Identities: Second De Morgan Law

Example: Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution: We prove this identity by showing that:

1) $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ and

2) $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

Continued on next slide →

Proving Set Identities: Second De Morgan Law (Cont.)

1) Show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$x \in \overline{A \cap B}$	by assumption
$x \notin A \cap B$	defn. of complement
$\neg((x \in A) \wedge (x \in B))$	defn. of intersection
$\neg(x \in A) \vee \neg(x \in B)$	1st De Morgan Law for Prop Logic
$x \notin A \vee x \notin B$	defn. of negation
$x \in \overline{A} \vee x \in \overline{B}$	defn. of complement
$x \in \overline{A} \cup \overline{B}$	defn. of union

Continued on next slide →

Proving Set Identities: Second De Morgan Law (Cont.)

2) Show that: $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

$x \in \overline{A} \cup \overline{B}$	by assumption
$(x \in \overline{A}) \vee (x \in \overline{B})$	defn. of union
$(x \notin A) \vee (x \notin B)$	defn. of complement
$\neg(x \in A) \vee \neg(x \in B)$	defn. of negation
$\neg((x \in A) \wedge (x \in B))$	by 1st De Morgan Law for Prop Logic
$\neg(x \in A \cap B)$	defn. of intersection
$x \in \overline{A \cap B}$	defn. of complement

Proving Set Identities with Propositional Logic

We can also prove the second De Morgan's law by the following process with propositional logic on Set Builder Notation:

$$
\begin{array}{rcll}
\overline{A \cap B} & = & \{x | x \notin A \cap B\} & \text{by defn. of complement} \\
& = & \{x | \neg(x \in (A \cap B))\} & \text{by defn. of does not belong symbol} \\
& = & \{x | \neg(x \in A \wedge x \in B\} & \text{by defn. of intersection} \\
& = & \{x | \neg(x \in A) \vee \neg(x \in B)\} & \text{by 1st De Morgan law} \\
& & & \text{for Prop Logic} \\
& = & \{x | x \notin A \vee x \notin B\} & \text{by defn. of not belong symbol} \\
& = & \{x | x \in \overline{A} \vee x \in \overline{B}\} & \text{by defn. of complement} \\
& = & \{x | x \in \overline{A} \cup \overline{B}\} & \text{by defn. of union} \\
& = & \overline{A} \cup \overline{B} & \text{by meaning of notation}
\end{array}
$$

Therefore,

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

Proving Set Identities with Membership Tables

We can use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$:

TABLE 2 **A Membership Table for the Distributive Property.**

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Lecture 8: Sequences & Summations

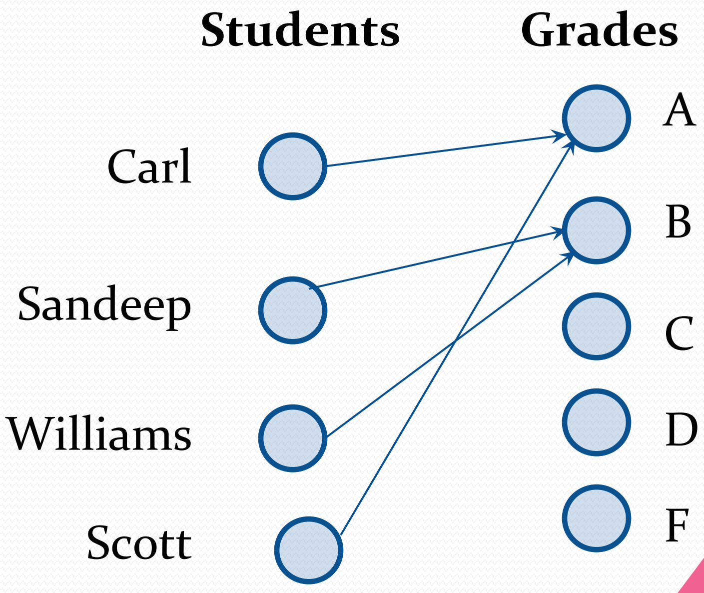
Section 2.4

Section Summary

- Sequences.
  - Examples: Geometric Progression, Arithmetic Progression
- Recurrence Relations
  - Example: Fibonacci Sequence
- Summations

Functions

Definition: Let A and B be nonempty sets. A *function* f from A to B, denoted $f : A \to B$ is an assignment of each element of A to exactly one element of B. We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A.

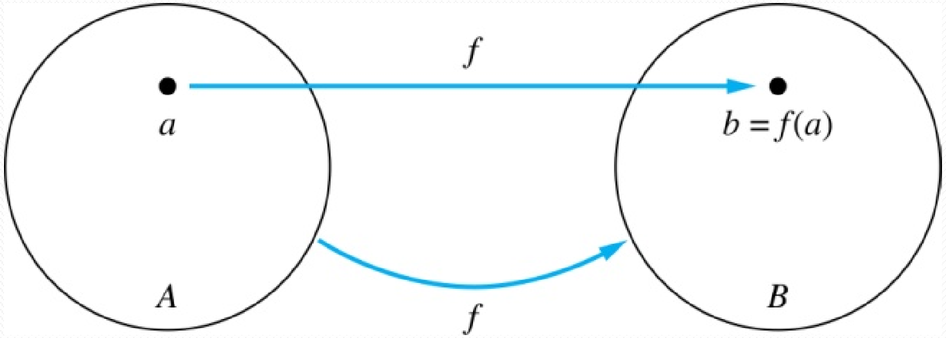


- Functions are sometimes called *mappings* or *transformations*.

Functions

Given a function $f: A \to B$:

- We say f *maps* A to B *or* f is a *mapping* from A to B.
- A is called the ***domain*** of f.
- B is called the ***codomain*** of f.
- If $f(a) = b$,
  - then b is called the ***image*** of a under f.
  - a is called the ***preimage*** of b.
- The ***range*** of f is the set of all images of points in **A** under f. We denote it by $f(\boldsymbol{A})$.



Sequences

- Sequences are **ordered lists** of elements.
  - 1, 2, 3, 5, 8   (finite)
  - 1, 3, 9, 27, 81, ……. (infinite)

Definition: A *sequence* is a **function** from a subset of the integers (usually either the set {0, 1, 2, 3, 4, …} or {1, 2, 3, 4, …}) to a set S.

- The notation a_n is used to denote the **image** of the integer n. We can think of a_n as the equivalent of $f(n)$ where f is a function from {0, 1, 2, …} to S. We call a_n a ***term*** of the sequence.

Sequences

Example: Consider the sequence $\{a_n\}$ where $a_n = \frac{1}{n}$

$$\{a_n\} = \{a_1, a_2, a_3, \ldots\}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \ldots$$

Geometric Progression

Definition: A *geometric progression* is a sequence of the form:

$$a, ar, ar^2, \ldots, ar^n, \ldots$$

where the *initial term a* and the *common ratio r* are real numbers.

Examples:

1. Let $a = 1$ and $r = -1$. Then:
$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \ldots\} = \{1, -1, 1, -1, 1, \ldots\}$$

1. Let $a = 2$ and $r = 5$. Then:
$$\{c_n\} = \{c_0, c_1, c_2, c_3, c_4, \ldots\} = \{2, 10, 50, 250, 1250, \ldots\}$$

1. Let $a = 6$ and $r = 1/3$. Then:
$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \ldots\} = \{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots\}$$

Arithmetic Progression

Definition: A *arithmetic progression* is a sequence of the form:

$$a, a + d, a + 2d, \dots, a + nd, \dots$$

where the *initial term a* and the *common difference d* are real numbers.

Examples:

1. Let $a = -1$ and $d = 4$: $\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11, 15, \dots\}$

1. Let $a = 7$ and $d = -3$: $\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$

1. Let $a = 1$ and d = 2: $\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$

Recurrence Relations

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the **previous terms of the sequence**, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

- A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.
- The *initial conditions* for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Recurrence Relations

Example 1: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1,2,3,4,\ldots.$ and suppose that $a_0 = 2$.

What are a_1 , a_2 and a_3?

[Here $a_0 = 2$ is the initial condition.]

Solution: We see from the recurrence relation that

$a_1 = a_0 + 3 = 2 + 3 = 5$

$a_2 = 5 + 3 = 8$

$a_3 = 8 + 3 = 11$

Recurrence Relations

Example 2: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2,3,4,\ldots.$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3?

[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.]

Solution: We see from the recurrence relation that

$$a_2 = a_1 - a_0 = 5 - 3 = 2$$

$$a_3 = a_2 - a_1 = 2 - 5 = -3$$

Fibonacci Sequence

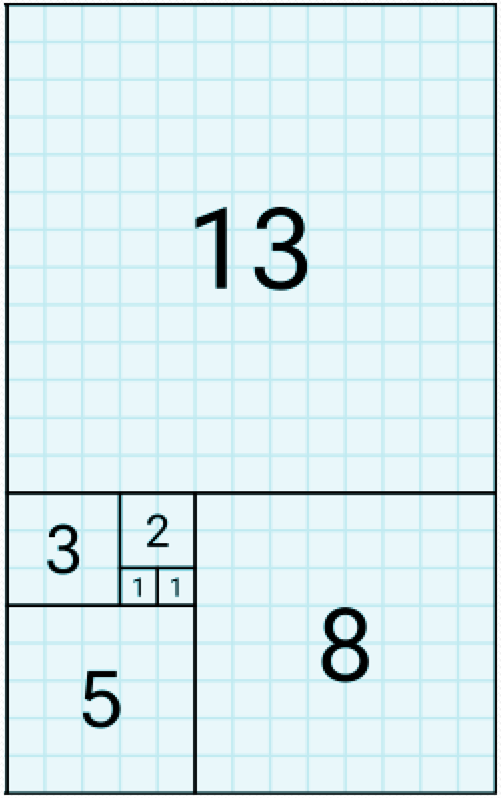
Definition: We define the *Fibonacci sequence*, $f_0, f_1, f_2, \ldots,$ by:

- Initial Conditions: $f_0 = 0, f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

Example: Find f_2, f_3, f_4, f_5 and f_6.

Answer:

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$
$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$
$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$
$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$
$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$



Solving Recurrence Relations

- Finding a **formula for the n^{th} term** of the sequence generated by a recurrence relation is called ***solving the recurrence relation***.
- Such a formula is called a *closed formula*.
- Various methods for solving recurrence relations will be covered in next lectures (Chapter 8) where recurrence relations will be studied in greater depth.
- For now, we illustrate by example the **method of iteration** in which we need to **guess** the formula. The guess can be proved correct by the method of induction.

Iterative Solution Example

Method 1: Working upward, forward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n\text{-}1} + 3$ for $n = 2, 3, 4, ...$ and suppose that $a_1 = 2$.

$$a_2 = 2 + 3$$

$$a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$$

$$a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$$

.

.

.

$$a_n = a_{n\text{-}1} + 3 = \mathbf{(2 + 3 \cdot (n - 2))} + 3 = 2 + 3(n - 1)$$

Iterative Solution Example

Method 2: Working downward, backward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \ldots.$ and suppose that $a_1 = 2$.

$$
\begin{aligned}
a_n &= a_{n-1} + 3 \\
&= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\
&= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \\
&\;\;\vdots \\
&= a_2 + 3(n-2) = (\boldsymbol{a_1} + 3) \mathbf{+ 3(n-2)} = 2 + 3(n-1)
\end{aligned}
$$

Real World Application

Example: Suppose that a person deposits \$10,000.00 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?

Solution:

Let P_n denote the amount in the account after n years.

Then, P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11 P_{n-1} = (1.11)\ P_{n-1}$$

with the initial condition $P_0 = 10{,}000$.

Continued on next slide ⇒

Real World Application

$$P_n = P_{n-1} + 0.11 P_{n-1} = (1.11)\ P_{n-1}$$

with the initial condition $P_0 = 10{,}000$

Solving the recurrence with Forward Substitution:

$P_1 = (1.11) P_0$

$P_2 = (1.11) P_1 = (1.11)^2 P_0$

$P_3 = (1.11) P_2 = (1.11)^3 P_0$

:

$P_n = (1.11) P_{n-1} = (1.11)^n P_0 = (1.11)^n 10{,}000$

Since, $P_n = (1.11)^n 10{,}000$, $P_{30} = (1.11)^{30}\ 10{,}000 = \$228{,}992.97$.

Identifying Integer Sequences

- Given a few terms of a sequence, try to identify the sequence. Conjecture a formula, recurrence relation, or some other rule.
- Some questions to ask?
  - Are there **repeated terms** of the same value?
  - Can you obtain a term from the previous term by **adding** an amount or **multiplying** by an amount?
  - Can you obtain a term by **combining the previous terms** in some way?
  - Are there cycles among the terms?
  - Do the terms match those of a well known sequence?

Identifying Integer Sequences

Find formulae for the sequences with the following first five terms:

Example 1: 1, 1/2, 1/4, 1/8, 1/16, ….

Solution: Note that the denominators are powers of 2.

The sequence with $a_n = 1/2^n$ is a possible match.

This is a **geometric** progression with $a = 1$ and $r = 1/2$.

Example 2: 1,3,5,7,9

Solution: Note that each term is obtained by adding 2 to the previous term. A possible formula is $a_n = 2n + 1$.

This is an **arithmetic** progression with $a = 1$ and $d = 2$.

Example 3: 1, –1, 1, –1,1

Solution: The terms alternate between 1 and -1.

A possible sequence is $a_n = (-1)^n$.

This is a **geometric** progression with $a = 1$ and $r = -1$.

Useful Sequences

TABLE 1 **Some Useful Sequences.**

nth Term	*First 10 Terms*
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, . . .
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, . . .
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, . . .
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, . . .
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, . . .
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, . . .

Guessing Sequences

Example: Conjecture a simple formula for a_n if the first 10 terms of the sequence $\{a_n\}$ are
1, 7, 25, 79, 241, 727, 2185, 6559, 19681, 59047.

Solution: Note the ratio of each term to the previous approximates 3. So now compare with the sequence 3^n.

$\{3^n\} = 1, 3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, \ldots$

We notice that the nth term is 2 less than the corresponding power of 3. So a good conjecture is that $a_n = 3^n - 2$.

Summations

- Sum of the terms $a_m, a_{m+1}, \ldots, a_n$ from the sequence $\{a_n\}$
- The notations:

$$\sum_{j=m}^{n} a_j \quad \text{or,} \quad \textstyle\sum_{j=m}^{n} a_j \quad \text{or,} \quad \textstyle\sum_{m \leq j \leq n} a_j$$

represents $$a_m + a_{m+1} + \cdots + a_n$$

- The variable *j* is called the *index of summation*.
It runs through all the integers starting with its *lower limit, m* and ending with its *upper limit, n*.

Summations

- More generally for a set S:

$$\sum_{j \in S} a_j$$

- **Examples**:

$$r^0 + r^1 + r^2 + r^3 + \cdots + r^n = \sum_0^n r^j$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \sum_1^\infty \frac{1}{i}$$

$$\text{If } S = \{2, 5, 7, 10\} \text{ then } \sum_{j \in S} a_j = a_2 + a_5 + a_7 + a_{10}$$

Summations Examples

1. What is the value of $\sum_{j=1}^{5} j^2$?

Solution:
$$\begin{aligned}\sum_{j=1}^{5} j^2 &= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 \\ &= 1 + 4 + 9 + 16 + 25 \\ &= 55.\end{aligned}$$

1. What is the value of $\sum_{k=4}^{8} (-1)^k$?

Solution:
$$\begin{aligned}\sum_{k=4}^{8} (-1)^k &= (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8 \\ &= 1 + (-1) + 1 + (-1) + 1 \\ &= 1.\end{aligned}$$

Summation of Series Formula

Sums of terms of geometric progressions

$$\sum_{j=0}^{n} ar^j = \begin{cases} \frac{ar^{n+1}-a}{r-1} & r \neq 1 \\ (n+1)a & r = 1 \end{cases}$$

Proof: Let $S_n = \sum_{j=0}^{n} ar^j$ To compute S_n, first multiply both sides of the equality by r and then manipulate the resulting sum:

$$rS_n = r\sum_{j=0}^{n} ar^j = \sum_{j=0}^{n} ar^{j+1}$$

Continued on next slide ⇒

Summation of Series Formula

$$= \sum_{j=0}^{n} ar^{j+1}$$ From previous slide.

$$= \sum_{k=1}^{n+1} ar^{k}$$ Shifting the index of summation with $k = j + 1$.

$$= \left(\sum_{k=0}^{n} ar^{k}\right) + (ar^{n+1} - a)$$ Removing $k = n + 1$ term and adding $k = 0$ term.

$$= S_n + (ar^{n+1} - a)$$ Substituting S for summation formula

$$\therefore \; rS_n = S_n + (ar^{n+1} - a)$$

$$S_n = \frac{ar^{n+1} - a}{r - 1}$$ if $r \neq 1$

$$S_n = \sum_{j=0}^{n} ar^{j} = \sum_{j=0}^{n} a = (n+1)a$$ if $r = 1$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.	
Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1}-a}{r-1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x\rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \lvert x\rvert < 1$	$\frac{1}{(1-x)^2}$

for Arithmetic progression, the sum of first n terms:

$$\sum_{i=0}^{n-1} (a + id) = \frac{n}{2}\{2a + (n-1)d\}$$

We can prove these by induction, also.

Proof requires calculus

Lecture 9: Solving Linear Recurrence Relations

Section 8.2

Section Summary

- Linear Homogeneous Recurrence Relations
- Solving Linear Homogeneous Recurrence Relations with Constant Coefficients.
- Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients. (Self Study)

Linear Homogeneous Recurrence Relations

Definition: A *linear homogeneous recurrence relation of degree* k *with constant coefficients* is a recurrence relation of the form $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots.. + c_k\, a_{n-k}$, where $c_1, c_2, \ldots., c_k$ are real numbers, and $c_k \neq 0$.

- It is *linear* because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of *n*.
- It is *homogeneous* because no terms occur that are not multiples of the recursive elements.
- Each coefficient is a constant.
- The *degree* is *k* because a_n is expressed in terms of the previous *k* terms of the sequence.

Examples of Linear Homogeneous Recurrence Relations

• $P_n = (1.11)P_{n-1}$	linear homogeneous recurrence relation of degree one
• $f_n = f_{n-1} + f_{n-2}$	linear homogeneous recurrence relation of degree two
• $a_n = a_{n-1} + a_{n-2}^2$	not linear
• $H_n = 2H_{n-1} + 1$	not homogeneous
• $B_n = nB_{n-1}$	coefficients are not constants
• $a_n = a_{n-5}$	linear homogeneous recurrence relation of degree **five**

Solving Linear Homogeneous Recurrence Relations

- The basic approach is to look for solutions of the form

$$a_n = r^n, \text{ where } r \text{ is a constant.}$$

- Notice that $a_n = r^n$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

if and only if $r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k}$.

- Divide both sides of this equation by r^{n-k}.

$$r^k = c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_{k-1} r + c_k$$

Solving Linear Homogeneous Recurrence Relations

- Algebraic manipulation yields the *characteristic equation*:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0$$

- The sequence $\{a_n\}$ with $a_n = r^n$ is a solution if and only if

  r is a solution to this equation.

- The solutions to the characteristic equation are called the *characteristic roots* of the recurrence relation. The roots are used to give an explicit formula for all the solutions of the recurrence relation.

Solving Linear Homogeneous Recurrence Relations

- Another observation:

A **linear combination** of two solutions is also a solution.

So, if the recurrence $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ have s_n and t_n both as valid solutions, so,

$$s_n = c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k} \quad \text{and}$$
$$t_n = c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}.$$

Consider a linear combination of s_n and t_n : $b_1 s_n + b_2 t_n$, where b_1 and b_2 are real numbers.

Solving Linear Homogeneous Recurrence Relations

We have:

$$
\begin{aligned}
&b_1 s_n + b_2 t_n \\
&= b_1(c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k}) + b_2(c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}) \\
&= c_1(b_1 s_{n-1} + b_2 t_{n-1}) + c_2(b_1 s_{n-2} + b_2 t_{n-2}) + \cdots + c_k(b_1 s_{n-k} + b_k t_{n-k}).
\end{aligned}
$$

- So, if s_n and t_n both are solutions, then $b_1 s_n + b_2 t_n$ is also a solution.
- Combining that with the idea solutions can be of the form r^n, gives us the following theorem:

Solving Linear Homogeneous Recurrence Relations of Degree Two

Let c_1 and c_2 be real numbers.

Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2.

Now, the degree-2 recurrence relation, $\boldsymbol{a_n = c_1 a_{n-1} + c_2 a_{n-2}}$ has the solution of the form: $\boldsymbol{a_n = \alpha_1 r_1{}^n + \alpha_2 r_2{}^n}$ for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

THEOREM 1 Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2. Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

Solving Linear Homogeneous Recurrence Relations of Degree Two

Example: What is the solution to the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2} \text{ with } a_0 = 2 \text{ and } a_1 = 7?$$

Solution:

- Step-1: find the *characteristic equation*

  For our degree 2 recurrence relations, the characteristic equation is $r^2 - r - 2 = 0$. ($c_1 = 1, c_2 = 2$)

- Step-2: find the *roots* of the characteristic equation

  Its roots are $r = 2$ and $r = -1$.

- Step-3: find the *general solution* format with α_1 and α_2

  Therefore, $\{a_n\}$ is a solution if and only if $\mathbf{a_n = \alpha_1 2^n + \alpha_2(-1)^n}$, for some constants α_1 and α_2.

- Step-4: find the values of α_1 and α_2 using **the initial conditions**

  $a_0 = \mathbf{2 = \alpha_1 + \alpha_2}$ and $a_1 = \mathbf{7 = \alpha_1 2 + \alpha_2(-1)}$.

  Solving these equations, we find that $\alpha_1 = 3$ and $\alpha_2 = -1$.

Hence, the solution is the sequence $\{a_n\}$ with $a_n = 3 \cdot 2^n - (-1)^n$.

An Explicit Formula for the Fibonacci Numbers

We can find an explicit formula for the Fibonacci numbers. The sequence of Fibonacci numbers satisfies the recurrence relation: $f_n = f_{n-1} + f_{n-2}$ with the initial conditions: $f_0 = 0$ and $f_1 = 1$.

Solution: The roots of the characteristic equation $r^2 - r - 1 = 0$ are:

$$r_1 = \frac{1+\sqrt{5}}{2}, \; r_2 = \frac{1-\sqrt{5}}{2}.$$

So, for some constants α_1 and α_2,

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

Using the initial conditions:

$$f_0 = \alpha_1 + \alpha_2 = 0; \qquad f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right) = 1$$

We solve these equations to find: $\alpha_1 = \frac{1}{\sqrt{5}}, \quad \alpha_2 = -\frac{1}{\sqrt{5}}$

Thus, $$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n$$

What if there is a repeated root from the characteristic equation?

Let c_1 and c_2 be real numbers with $c_2 \neq 0$.

Suppose that $r^2 - c_1 r - c_2 = 0$ has one repeated root r_0.

Then the sequence $\{a_n\}$ is a solution to the recurrence relation $\boldsymbol{a_n = c_1 a_{n-1} + c_2 a_{n-2}}$ if and only if

$$\boldsymbol{a_n = \alpha_1 r_0^{\,n} + \alpha_2 \mathrm{n} r_0^{\,n}}$$

for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

THEOREM 2 Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0. A sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$, for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

Example Problem

Example: What is the solution to the recurrence relation

$a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$?

Solution: The characteristic equation is $r^2 - 6r + 9 = 0$.

The only root is $r = 3$.

Therefore, $\{a_n\}$ is a solution if and only if $\boldsymbol{a_n = \alpha_1 3^n + \alpha_2 n(3)^n}$,

where α_1 and α_2 are constants.

From the given initial conditions,

$a_0 = 1 = \alpha_1$ and, $a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 1 \cdot 3$.

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = 1$.

Hence, $a_n = 3^n + n3^n$.

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

This theorem can be used to solve linear homogeneous recurrence relations with constant coefficients of any degree when the characteristic equation has distinct roots.

Theorem 3: Let c_1, c_2 ,…, c_k be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \cdots - c_k = 0$$

has k distinct roots r_1, r_2, …, r_k. Then a sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots + c_k a_{n-k}$
if and only if

$$a_n = \alpha_1 r_1{}^n + \alpha_2 r_2{}^n + \ldots + \alpha_k r_k{}^n.$$

for $n = 0, 1, 2, \ldots$, where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are constants.

Example Problem

Example: What is the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3} \text{ with } a_0 = 2, a_1 = 5, a_2 = 15?$$

Solution: The characteristic equation is $\boldsymbol{r^3 - 6r^2 + 11r - 6 = 0}$.

The roots are, $r_1 = 1$, $r_2 = 2$, $r_3 = 3$.

Therefore, the solution is of the form $\boldsymbol{a_n = \alpha_1 1^n + \alpha_2 2^n + \alpha_3 3^n}$,

where α_1, α_2 and α_3 are constants.

From the given initial conditions,

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3 \text{ , } a_1 = 5 = \alpha_1 + \alpha_2 \cdot 2 + \alpha_3 \cdot 3 \text{ and}$$

$$a_2 = 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9$$

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = -1$ and $\alpha_3 = 2$.

Hence, $a_n = 1 - 2^n + 2 \cdot 3^n$.

The General Case with Repeated Roots Allowed

Theorem 4: Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \cdots - c_k = 0$$

has t distinct roots $r_1, r_2, \ldots, r_t$ with multiplicities $m_1, m_2, \ldots, m_t$, respectively so that $m_i \geq 1$ for $i = 0, 1, 2, \ldots, t$ and $m_1 + m_2 + \ldots + m_t = k$.

Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = {} & (\alpha_{1,0} + \alpha_{1,1} n + \cdots + \alpha_{1,m_1-1} n^{m_1-1}) r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1} n + \cdots + \alpha_{2,m_2-1} n^{m_2-1}) r_2^n \\ & + \cdots + (\alpha_{t,0} + \alpha_{t,1} n + \cdots + \alpha_{t,m_t-1} n^{m_t-1}) r_t^n \end{aligned}$$

for $n = 0, 1, 2, \ldots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_{i-1}$.

Example Problem

Example: Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5 and 9. What is the form of the general solution?

Solution: Here, there are only 3 roots,

$r_1 = 2$, with multiplicity 3, $r_2 = 5$, with multiplicity 2 and

$r_3 = 9$, with multiplicity 1.

Therefore, the solution is of the form:

$$a_n = (\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2)(2)^n$$
$$+ (\alpha_{2,0} + \alpha_{2,1}n)(5)^n$$
$$+ (\alpha_{3,0})(9)^n$$

Example Problem

Example: What is the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3} \text{ with } a_0 = 1, a_1 = -2, a_2 = -1?$$

Solution: The characteristic equation is $\boldsymbol{r^3 + 3r^2 + 3r + 1 = 0}$.

There is only a single root, $r = -1$, with multiplicity 3.

Therefore, the solution is of the form

$$\boldsymbol{a_n = \alpha_{1,0}(-1)^n + \alpha_{1,1}n(-1)^n + \alpha_{1,2}n^2(-1)^n}$$

$$\boldsymbol{a_n = (\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2)(-1)^n}$$

where $\alpha_{1,0}$, $\alpha_{1,1}$ and $\alpha_{1,2}$ are constants.

From the given initial conditions,

$$a_0 = 1 = \alpha_{1,0}\text{ , } a_1 = -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,2} \text{ and}$$

$$a_2 = -1 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,2}.$$

Solving, we find that $\alpha_{1,0} = 1$ and $\alpha_{1,1} = 3$ and $\alpha_{1,2} = -2$.

Hence, $a_n = (1 + 3n - 2n^2)(-1)^n$.

Lecture 10: The Basic Rules of Counting

Section 6.1

Lecture Summary

- The Product Rule
- The Sum Rule
- The Subtraction Rule

Basic Counting Principles: The Product Rule

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

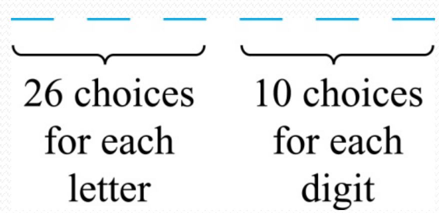
Example: How many bit strings of length seven are there?

Solution: Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.

The Product Rule

Example: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?

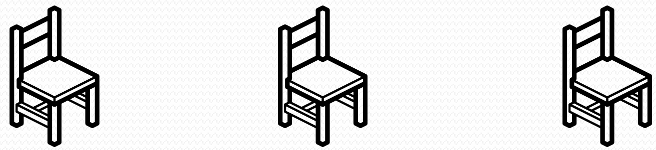
Solution: By the product rule,
there are $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17{,}576{,}000$ different possible license plates.



Seating Arrangement

Example: We have 5 students, how many different seating arrangements are possible if we have 3 seats available?

Possible solution:



For the first chair, we have 5 students available to choose from

For the second chair, we have 4 students available now to choose from

For the third chair, we have remaining 3 students to choose from

and all three events have to occur all at once, so we can use product rule.
And possible arrangement would be $(5 \times 4 \times 3) = 60$

We can later try to solve the same problem with permutation or combination concept.

Telephone Numbering Plan

Example: The *North American numbering plan* (*NANP*) specifies that a telephone number consists of 10 digits, consisting of a three-digit area code, a three-digit office code, and a four-digit station code. There are some restrictions on the digits.

- Let X denote a digit from 0 through 9.
- Let N denote a digit from 2 through 9.
- Let Y denote a digit that is 0 or 1.

- In the old plan (in use in the 1960s) the format was *NYX-NNX-XXXX*.
- In the new plan, the format is *NXX-NXX-XXXX*.

How many different telephone numbers are possible under the old plan and the new plan?

Solution: Use the Product Rule.

- There are $8 \cdot 2 \cdot 10 = 160$ area codes with the format *NYX.*
- There are $8 \cdot 10 \cdot 10 = 800$ area codes with the format *NXX.*
- There are $8 \cdot 8 \cdot 10 = 640$ office codes with the format *NNX.*
- There are $10 \cdot 10 \cdot 10 \cdot 10 = 10{,}000$ station codes with the format *XXXX.*
  - Number of old plan telephone numbers: $160 \cdot 640 \cdot 10{,}000 = 1{,}024{,}000{,}000$.
  - Number of new plan telephone numbers: $800 \cdot 800 \cdot 10{,}000 = 6{,}400{,}000{,}000$.

Counting Subsets of a Finite Set

Counting Subsets of a Finite Set: Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

Solution:
When the elements of S are listed in an arbitrary order, there is a one-to-one correspondence between **subsets of S** and **bit strings of length $|S|$**, when the ith element is in the subset, the bit string has a 1 in the ith position and a 0 otherwise.

By the product rule, there are $2^{|S|}$ such bit strings, and therefore $2^{|S|}$ subsets.

Product Rule in Terms of Sets

- If $A_1, A_2, \ldots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements of each set.
- The task of choosing an element in the Cartesian product $A_1 \times A_2 \times \cdots \times A_m$ is done by choosing an element in A_1, an element in A_2, …, and an element in A_m.
- By the product rule, it follows that:
  $|A_1 \times A_2 \times \cdots \times A_m| = |A_1| \cdot |A_2| \cdot \cdots \cdot |A_m|.$

Basic Counting Principles: The Sum Rule

The Sum Rule: If a task can be done **either** in one of n_1 ways **or** in one of n_2, where none of the set of n_1 ways is the **same** as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick a representative.

The Sum Rule in terms of sets

- The sum rule can be phrased in terms of sets.

  $|A \cup B| = |A| + |B|$ as long as A and B are disjoint sets.
- Or more generally,

  $$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

  when $A_i \cap A_j = \emptyset$ for all i, j.

- The case where the sets have elements in common will be discussed when we consider the subtraction rule (remember inclusion-exclusion principle?)

Combining the Sum and Product Rule

Example: Suppose variable names in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible names.

Solution: Two ways to make a label.

Only one letter = 26,

1 Letter followed by a digit = 26*10 = 260 *[product rule]*

Total labels = 26 + 260 = 286 *[sum rule]*

Example: Counting Passwords

Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution: Let P be the total number of passwords, and let P_6, P_7, and P_8 be the passwords of length 6, 7, and 8.

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6, P_7, and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:
  - $P_6 = 36^6 - 26^6 = 2{,}176{,}782{,}336 - 308{,}915{,}776 = 1{,}867{,}866{,}560.$
  - $P_7 = 36^7 - 26^7 = 78{,}364{,}164{,}096 - 8{,}031{,}810{,}176 = 70{,}332{,}353{,}920.$
  - $P_8 = 36^8 - 26^8 = 2{,}821{,}109{,}907{,}456 - 208{,}827{,}064{,}576$
    $= 2{,}612{,}282{,}842{,}880.$
- Consequently, $P = P_6 + P_7 + P_8 = 2{,}684{,}483{,}063{,}360.$

Internet Addresses

❖ Version 4 of the Internet Protocol (IPv4) uses 32 bits.

Bit Number	0	1	2	3	4	8	16	24	31
Class A	0	netid				hostid			
Class B	1	0	netid				hostid		
Class C	1	1	0	netid				hostid	
Class D	1	1	1	0	Multicast Address				
Class E	1	1	1	1	0	Address			

❖ **Class A Addresses**: used for the largest networks, a 0, followed by a 7-bit net_id and a 24-bit host_id.

❖ **Class B Addresses**: used for the medium-sized networks, a 10, followed by a 14-bit net_id and a 16-bit host_id.

❖ **Class C Addresses**: used for the smallest networks, a 110, followed by a 21-bit net_id and a 8-bit host_id.

- ➢ Neither Class D nor Class E addresses are assigned as the address of a computer on the internet. Only Classes A, B, and C are available.
- ➢ 1111111 is not available as the net_id of a Class A network.
- ➢ Host_ids consisting of all 0s and all 1s are not available in any network.

Counting Internet Addresses

Example: How many different IPv4 addresses are available for computers on the internet?

Solution: Use both the sum and the product rule. Let x be the number of available addresses, and let x_A, x_B, and x_C denote the number of addresses for the respective classes.

- To find, x_A: $2^7 - 1 = 127$ net_ids. $2^{24} - 2 = 16{,}777{,}214$ host_ids.

  $x_A = 127 \cdot 16{,}777{,}214 = 2{,}130{,}706{,}178$.
- To find, x_B: $2^{14} = 16{,}384$ net_ids. $2^{16} - 2 = 16{,}534$ host_ids.

  $x_B = 16{,}384 \cdot 16{,}534 = 1{,}073{,}709{,}056$.
- To find, x_C: $2^{21} = 2{,}097{,}152$ net_ids. $2^8 - 2 = 254$ host_ids.

  $x_C = 2{,}097{,}152 \cdot 254 = 532{,}676{,}608$.
- Hence, the total number of available IPv4 addresses is

  $$x = x_A + x_B + x_C$$
  $$= 2{,}130{,}706{,}178 + 1{,}073{,}709{,}056 + 532{,}676{,}608$$
  $$= 3{,}737{,}091{,}842.$$

Not Enough Today !!
The newer IPv6 protocol solves the problem of too few addresses.

Basic Counting Principles: Subtraction Rule

Subtraction Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

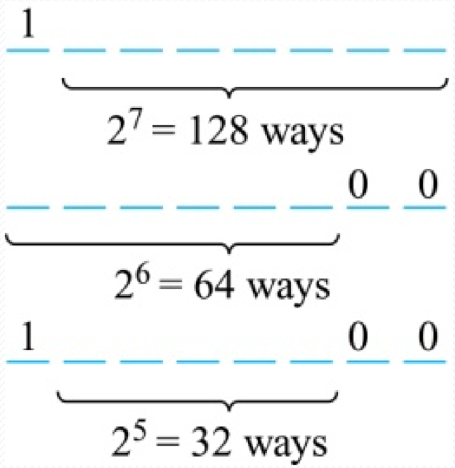
- Also known as, the *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Counting Bit Strings

Example: How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution: Use the subtraction rule.



- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that end with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$
- Hence, the number is $128 + 64 - 32 = 160$.

Lecture 11: The Pigeonhole Principle

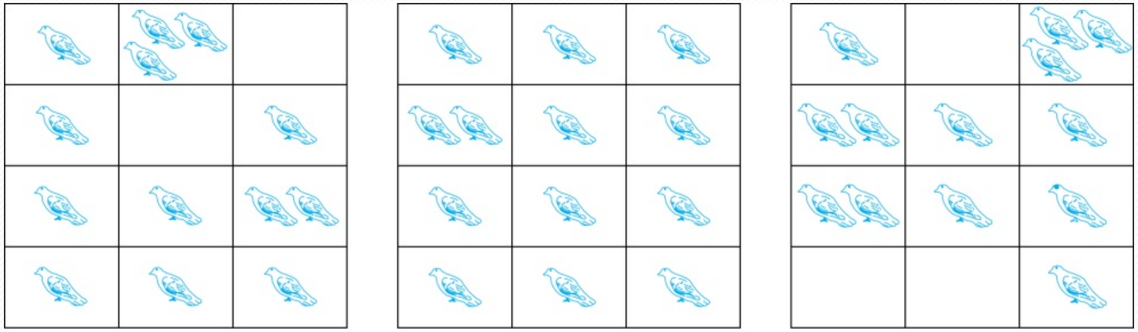
Section 6.2

Lecture Summary

- The Pigeonhole Principle
- The Generalized Pigeonhole Principle

The Pigeonhole Principle

If a flock of 13 pigeons roosts in a set of 12 pigeonholes, one of the pigeonholes must have more than 1 pigeon.



(a) (b) (c)

Pigeonhole Principle: If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Proof: We use a proof by contraposition. Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k. This contradicts the statement that we have $k + 1$ objects.

Picking Object Blindfolded

Example: Suppose you are blindfolded and picking out balls from a box like below.

(a) Minimum how many balls to pick to get 2 balls of same color?

(b) Minimum how many balls to pick to get 2 red balls?

8 Red
5 blue
6 Black

Picking Object Blindfolded

Example: Suppose you are blindfolded and picking out balls from a box like below.

(a) Minimum how many balls to pick to get 2 balls of same color?

(b) Minimum how many balls to pick to get 2 red balls?

8 Red
5 blue
6 Black

(a) Worst case scenario would be, you pick 1 balls from each color = 3 balls. Then if you pick your 4th ball, it must match any of the color, and you will have your 2 same color balls.
So, solution is (1 Red+1 Blue + 1 Black) + 1 = 4 balls

(b) Worst case scenario would be, you pick all the blue and black balls. Then if you pick your next 2 balls, it must be of red color.
So, solution is (5 Blue + 6 Black + 1 Red) + 1 = 13 balls

Pigeonhole Principle

Example 1: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example 2: How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Solution: There are 101 possible scores on the final. The pigeonhole principle shows that among any 102 students there must be at least 2 students with the same score.

Pigeonhole Principle

Example 3: Show that for every integer n there is a multiple of n that has only 0s and 1s in its decimal expansion.

Solution: Let n be a positive integer. Consider the $n + 1$ integers 1, 11, 111, …., 11…1 (where the last has $n + 1$ 1s). There are n possible remainders when an integer is divided by n. By the pigeonhole principle, when each of the $n + 1$ integers is divided by n, at least two must have the same remainder. Subtract the smaller from the larger and the result is a multiple of n that has only 0s and 1s in its decimal expansion.

The Generalized Pigeonhole Principle

The Generalized Pigeonhole Principle: If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Proof: We use a proof by contraposition. Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects is at most

$$k\left(\left\lceil \frac{N}{k} \right\rceil - 1\right) < k\left(\left(\frac{N}{k} + 1\right) - 1\right) = N,$$

where the inequality $\lceil N/k \rceil < \lceil N/k \rceil + 1$ has been used. This is a contradiction because there are a total of n objects. ◀

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

The Generalized Pigeonhole Principle

Example: a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

b) How many must be selected to guarantee that at least three hearts are selected?

Solution: a) We assume four boxes; one for each suit. Using the generalized pigeonhole principle, at least one box contains at least $\lceil N/4 \rceil$ cards. At least three cards of one suit are selected if $\lceil N/4 \rceil \geq 3$. The smallest integer N such that $\lceil N/4 \rceil \geq 3$ is $N = 2 \cdot 4 + 1 = 9$.

b) A deck contains 13 hearts and 39 cards which are not hearts. So, if we select 41 cards, we may have 39 cards which are not hearts along with 2 hearts. However, when we select 42 cards, we must have at least three hearts. (Note that the generalized pigeonhole principle is **not** used here.)

Applications of Pigeonhole Principle

Example: During a month with 30 days, a baseball team plays at least one game a day, but no more than 45 games in total. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games in sum.

Solution: : Let a_j be the number of games played on or before the j-th day of the month. Then $a_1, a_2, \ldots, a_{30}$ is an increasing sequence of distinct positive integers, with $1 \leq a_j \leq 45$. Moreover, $a_1 + 14, a_2 + 14, \ldots, a_{30} + 14$ is also an increasing sequence of distinct positive integers, with $15 \leq a_j + 14 \leq 59$. The 60 positive integers $a_1, a_2, \ldots, a_{30}, a_1 + 14, a_2 + 14, \ldots, a_{30} + 14$ are all less than or equal to 59.

Hence, by the pigeonhole principle two of these integers are equal. Because the integers a_j , $j = 1, 2, \ldots, 30$ are all distinct and the integers $a_j + 14$, $j = 1, 2, \ldots, 30$ are all distinct, there must be indices i and j with $a_i = a_j + 14$. This means that exactly 14 games were played from day $j + 1$ to day i.

Lecture 12: Permutations & Combinations

Section 6.3 + 6.5

Section Summary

- Permutations
- Combinations
- Permutations with Repetitions
- Combinations with Repetitions

Permutations

Definition: A *permutation* of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of r elements of a set is called an *r-permutation*.

Example: Let $S = \{1,2,3\}$.

- The ordered arrangement 3,1,2 is a permutation of S.
- The ordered arrangement 3,2 is a 2-permutation of S.

- The number of r-permutations of a set with n elements is denoted by $P(n,r)$.
  - The 2-permutations of $S = \{1,2,3\}$ are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence, $P(3,2) = 6$.

Solving Counting Problems by Counting Permutations (*continued*)

Example: Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution: The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

A Formula for the Number of Permutations

Theorem: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r-permutations of a set with n distinct elements.

Proof: Use the product rule. The first element can be chosen in n ways. The second in $n - 1$ ways, and so on until there are $(n - (r - 1))$ ways to choose the last element.

- Note that $P(n, 0) = 1$, since there is only one way to order zero elements.

Corollary: If n and r are integers with $1 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Solving Counting Problems by Counting Permutations

Example: How many ways are there to select a first-prize winner, a second prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution:

$$P(100,3) = 100 \cdot 99 \cdot 98 = 970{,}200$$

Solving Counting Problems by Counting Permutations (*continued*)

Example: How many permutations of the letters *ABCDEFGH* contain the string *ABC*?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$P(6, 6) = 6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Combinations

Definition: An *r-combination* of elements of a set is an unordered selection of r elements from the set. Thus, an r-combination is simply a subset of the set with r elements.

- The number of r-combinations of a set with n distinct elements is denoted by $C(n, r)$. The notation $\binom{n}{r}$ is also used and is called a *binomial coefficient*.

Example: Let S be the set $\{a, b, c, d\}$. Then $\{a, c, d\}$ is a 3-combination from S. It is the same as $\{d, c, a\}$ since the order listed does not matter.

- $C(4,2) = 6$ because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Combinations

Theorem: The number of r-combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n,r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n,r) \cdot P(r,r)$. Therefore,

$$C(n,r) = \frac{P(n,r)}{P(r,r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}.$$

Combinations

Example: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$C(52,5) = \frac{52!}{5!47!}$$
$$= \frac{52\cdot 51\cdot 50\cdot 49\cdot 48}{5\cdot 4\cdot 3\cdot 2\cdot 1} = 26\cdot 17\cdot 10\cdot 49\cdot 12 = 2,598,960$$

- The different ways to select 47 cards from 52 is

$$C(52,47) = \frac{52!}{47!5!} = C(52,5) = 2,598,960.$$

This is a special case of a general result. →

Combinations

Corollary: Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n - r)$.

Proof: From Theorem 2, it follows that

$$C(n,r) = \frac{n!}{(n-r)!r!}$$

and $C(n,n-r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!}$.

Hence, $C(n, r) = C(n, n - r)$. ◀

This result can be proved without using algebraic manipulation. →

Combinations

Example: How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school.

Solution: The number of combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

Example: A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

Solution: The number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775\ .$$

Permutations with Repetition

Theorem: The number of r-permutations of a set of n objects with repetition allowed is n^r.

Proof: There are n ways to select an element of the set for each of the r positions in the r-permutation when repetition is allowed. Hence, by the product rule there are n^r r-permutations with repetition. ◀

Example: How many strings of length r can be formed from the uppercase letters of the English alphabet?

Solution: The number of such strings is 26^r, which is the number of r-permutations of a set with 26 elements.

Dealing with Constraints

Example: How many bit strings of length 10 contain
a) exactly 4 1s? b) At most 3 1s?

a) Solution: To create a bit string with exactly 4 ones, we need to choose 4 positions out of the 10 for the 1s, while the remaining 6 positions will automatically be 0s.
The number of ways to choose 4 positions out of 10 is: $C(10, 4) = 210$.

b) Solution: We need to consider multiple cases:
Case 1: Exactly 0 ones: This corresponds to a string where all 10 bits are 0.
There is only 1 such string.
Case 2: Exactly 1 one: Choose 1 position for the 1 out of 10.
The number of ways is: $C(10, 1)=10$
Case 3: Exactly 2 ones: Choose 2 positions for the 1s out of 10.
The number of ways is: $C(10, 2)=45$
Case 4: Exactly 3 ones: Choose 3 positions for the 1s out of 10.
The number of ways is: $C(10, 3)=120$
Total: Now, we use the sum rule for all these values: $1+10+45+120=176$

Dealing with Constraints

Example: 7 female and 9 male faculties are there in the mathematics department of a school. How many ways are there to select a committee of 5 members from the department if at least 1 female faculty is a must on the committee?

Solution: The total number of faculty members is 7 female and 9 male, for a total of 7+9=16. The number of ways to choose 5 members from 16 $= C(16,5) = 4368$.

Now we need to subtract the number of committees that contain no female, or committees that contain only male. Since there are 9 male, the number of ways to choose 5 members from 9 $= C(9, 5) = 126$.

Committees with at least 1 female $= 4368 - 126 = 4242$.

Dealing with Constraints

Example: How many strings of 6 lowercase letters of the English alphabet contain exactly 2 vowels?

Solution: First, we choose 2 positions for the vowels. The number of ways to do this is: $C(6,2) = 15$.

Each vowel position can be filled with any of the 5 vowels. Therefore, there are: $5^2 = 25$ ways to choose the 2 vowels.

The remaining 4 positions must be filled with consonants. Each position can be filled with any of the 21 consonants, so there are: $21^4 = 194\,481$ ways to choose the 4 consonants.

Finally, we apply product rule to get number of all the strings. Total strings $= 15 \times 25 \times 194\,481 = 72\,930\,375$

Permutations with Indistinguishable Objects

Example: How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution: There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420.$$

Lecture 13: Divisibility and Modular Arithmetic

Section 4.1

Section Summary

- Division
- Division Algorithm
- Modular Arithmetic

Division

Definition: If a and b are integers with $a \neq 0$, then a *divides* b if there exists an integer c such that $b = ac$.

- When a divides b we say that a is a *factor* or *divisor* of b and that b is a multiple of a.
- The notation $a \mid b$ denotes that a divides b.
- If $a \mid b$, then b/a is an integer.
- If a does not divide b, we write $a \nmid b$.

Example: Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Properties of Divisibility

Theorem 1: Let a, b, and c be integers, where $a \neq 0$.

i. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

ii. If $a \mid b$, then $a \mid bc$ for all integers c;

iii. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof: (i) Suppose $a \mid b$ and $a \mid c$, then it follows that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t). \quad \text{Hence, } a \mid (b + c) \quad \blacktriangleleft$$

Corollary: If a, b, and c be integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

Division Algorithm

When an integer is divided by a positive integer, there is a quotient and a remainder. This is traditionally called the "Division Algorithm," but is really a theorem.

Division Algorithm: If a is an integer and d a positive integer, then there are unique integers q and r, with $0 \leq r < d$, such that $a = dq + r$.

- d is called the *divisor*.
- a is called the *dividend*.
- q is called the *quotient*.
- r is called the *remainder*.

Definitions of Functions **div** and **mod**

$q = a \textbf{ div } d$
$r = a \textbf{ mod } d$

Division Algorithm

Since, $a = dq + r$ and $q = a \textbf{ div } d$ => $q = \lfloor a/d \rfloor$,

$r = a - dq$ => $r = a - d * \lfloor a/d \rfloor$

Examples:

- What are the quotient and remainder when 101 is divided by 11?

  Solution: The quotient when 101 is divided by 11 is $9 = 101 \textbf{ div } 11$, and the remainder is $2 = 101 \textbf{ mod } 11$.

- What are the quotient and remainder when -11 is divided by 3?

  Solution: The quotient when -11 is divided by 3 is $-4 = -11 \textbf{ div } 3$, and the remainder is $1 = -11 \textbf{ mod } 3$.

Division Algorithm

- Prove that if a is an integer that is not divisible by 3, then $(a+1)(a+2)$ is divisible by 3.

  Solution: Assume that $3 \nmid a$.

  So, we get a non-zero remainder, $r \in \{1, 2\}$ when a is divided by 3. In other words, $a = 3q + 1$ or, $a = 3q + 2$.

  Where $q \in \mathbf{Z}$. Now,

  Case 1: $a = 3q + 1$. Then, $(a+1)(a+2) = (3q+2)(3q+3)$ $= 3(3q+2)(q+1)$. Therefore, $3 \mid (a+1)(a+2)$.

  Case 2: $a = 3q + 2$. Then, $(a+1)(a+2) = (3q+3)(3q+4)$ $= 3(q+1)(3q+4)$. Therefore, $3 \mid (a+1)(a+2)$.

Congruence Relation

Definition: If a and b are integers and m is a positive integer, then a is *congruent* to b *modulo* m if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m.
- We say that $a \equiv b \pmod{m}$ is a *congruence* and that m is its *modulus.*
- Two integers are congruent mod m if and only if they have the same remainder when divided by m.
- If a is not congruent to b modulo m, we write

  $$a \not\equiv b \pmod{m}$$

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution:

- $17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.
- $24 \not\equiv 14 \pmod{6}$ since $24 - 14 = 10$ is not divisible by 6.

More on Congruences

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

- If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$.
- Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.
- Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, $m \mid a - b$ and $a \equiv b \pmod{m}$. ◀

The Relationship between (mod m) and **mod** m Notations

- The use of "mod" in $a \equiv b \pmod{m}$ and a **mod** $m = b$ are different.
  - $a \equiv b \pmod{m}$ is a **relation** on the set of integers.
  - In a **mod** $m = b$, the notation **mod** denotes a **function**.
- The relationship between these notations is made clear in this theorem.

Theorem 3: Let a and b be integers, and let m be a positive integer.

Then $a \equiv b \pmod{m}$ if and only if a **mod** $m = b$ **mod** m. (*Proof in the exercises*)

Congruences of Sums and Products

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof:

- Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.
- Therefore,
  - $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and
  - $b \cdot d = (a + sm)(c + tm) = ac + m(at + cs + stm)$.
- Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$. ◀

Example: Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Algebraic Manipulation of Congruences

- **Multiplying** both sides of a valid congruence by an integer preserves validity.

  If $a \equiv b \pmod{m}$ holds then $c \cdot a \equiv c \cdot b \pmod{m}$, where c is any integer, (Theorem 5 with $d = c$)

- **Adding** an integer to both sides of a valid congruence preserves validity.

  If $a \equiv b \pmod{m}$ holds then $c + a \equiv c + b \pmod{m}$, where c is any integer, (Theorem 5 with $d = c$)

- **Dividing** a congruence by an integer does **not** always produce a valid congruence.

Example: The congruence $14 \equiv 8 \pmod{6}$ holds. But dividing both sides by 2 does not produce a valid congruence since $14/2 = 7$ and $8/2 = 4$, but $7 \not\equiv 4 \pmod{6}$.

Computing the **mod** *m* Function of Products and Sums

We use the following corollary to Theorem 5 to compute the **remainder of the product or sum of two integers** when divided by m **from the remainders** when each is divided by m.

Corollary: Let m be a positive integer and let a and b be integers. Then

$$(a + b) \ (\mathbf{mod}\ m) = ((a\ \mathbf{mod}\ m) + (b\ \mathbf{mod}\ m))\ \mathbf{mod}\ m$$

and $ab\ \mathbf{mod}\ m = ((a\ \mathbf{mod}\ m)\ (b\ \mathbf{mod}\ m))\ \mathbf{mod}\ m$.

(*proof in textbook*)

Lecture 14: Primes and Greatest Common Divisors

Section 4.3

Section Summary

- Prime Numbers and their Properties
- Conjectures and Open Problems About Primes
- Greatest Common Divisors and Least Common Multiples
- The Euclidean Algorithm
- gcds as Linear Combinations

Primes

Definition: A positive integer p greater than 1 is called *prime* if the only positive factors of p are 1 and p. A positive integer that is greater than 1 and is not prime is called *composite*.

Example: The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.

The Fundamental Theorem of Arithmetic

Theorem: Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the **prime factors** are written in order of nondecreasing size.

Examples:

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$
- $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

Finding Prime Numbers

If an integer n is a composite integer,
then it has a prime divisor less than or equal to $\sqrt{n}$.

To see this, note that if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Trial division, a very inefficient method of determining if a number n is prime, is to try every integer $i \leq \sqrt{n}$ and see if n is divisible by i.

Sieve of Eratosthenes uses the trial division idea from the other way around.

continued →

Eratosthenes
(276-194 B.C.)

The Sieve of Eratosthenes

The *Sieve of Eratosthenes* can be used to find all primes not exceeding a specified positive integer. For example, begin with the list of integers between 1 and 100.

a. Delete all the integers, other than 2, divisible by 2.
b. Delete all the integers, other than 3, divisible by 3.
c. Next, delete all the integers, other than 5, divisible by 5.
d. Next, delete all the integers, other than 7, divisible by 7.
e. Since all the remaining integers are not divisible by any of the previous integers, other than 1, the primes are: {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}

continued →

Eratosthenes
(276-194 B.C.)

The Sieve of Eratosthenes

TABLE 1 **The Sieve of Eratosthenes.**

Integers divisible by 2 other than 2 receive an underline.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Integers divisible by 3 other than 3 receive an underline.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Integers divisible by 5 other than 5 receive an underline.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Integers divisible by 7 other than 7 receive an underline; integers in color are prime.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Since, if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

For determining the prime numbers within 100, we only need to divide with primes that do not exceed $\sqrt{100} = 10$. Meaning simply checking 2, 3, 5 and 7 is enough to find all primes under 100.

Study of Primes

- **Distribution of primes:** The number of primes not exceeding x, can be approximated by $\boldsymbol{x/\ln x}$. (The Prime Number Theorem)
- **Generating primes:** A closed formula that always produces primes has not been found so far. It is proven that there is no **polynomial** with integer coefficients such that $f(n)$ is prime for all positive integers n.
- **Conjectures on primes:**
    - **Goldbach's Conjecture:** Every even integer n, $n > 2$, is the sum of two primes.
    - There are infinitely many primes of the form $n^2 + 1$, where n is a positive integer. But it has been shown that there are infinitely many primes of the form $n^2 + 1$, where n is a positive integer or the product of at most two primes.
    - **The Twin Prime Conjecture:** There are infinitely many pairs of twin primes. Twin primes are pairs of primes that differ by 2. Examples are 3 and 5, 5 and 7, 11 and 13, etc.

Greatest Common Divisor

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the greatest common divisor of a and b. The greatest common divisor of a and b is denoted by gcd(a,b).

One can find greatest common divisors of small numbers by inspection.

Example: What is the greatest common divisor of 24 and 36?

Solution: $\gcd(24, 36) = 12$

Example: What is the greatest common divisor of 17 and 22?

Solution: $\gcd(17,22) = 1$

Co-primes / Relatively Primes

Definition: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: 17 and 22

Definition: The integers $a_1, a_2, \ldots, a_n$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution: Because $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$, we can say that 10, 17, and 21 are pairwise relatively prime.

Example: Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution: Because $\gcd(10, 24) = 2$, the given numbers are not pairwise relatively prime.

Finding the Greatest Common Divisor Using Prime Factorizations

- Suppose the prime factorizations of a and b are:

$$a = p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n}\ , \qquad b = p_1^{b_1} p_2^{b_2} \ldots p_n^{b_n}\ ,$$

  where each exponent is a nonnegative integer, and where all primes occurring in either prime factorization are included in both. Then:

$$\gcd(a,b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \ldots p_n^{\min(a_n,b_n)}\ .$$

- **Example**: $120 = 2^3 \cdot 3 \cdot 5 \qquad 500 = 2^2 \cdot 5^3$

  $\gcd(120{,}500) = 2^{\min(3,2)} \cdot 3^{\min(1,0)} \cdot 5^{\min(1,3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20$

- Finding the gcd of two positive integers using their prime factorizations is not efficient because there is no efficient algorithm for finding the prime factorization of a positive integer.

Least Common Multiple

Definition: The least common multiple of the positive integers *a* and *b* is the smallest positive integer that is divisible by both *a* and *b*. It is denoted by lcm(*a*,*b*).

- The least common multiple can also be computed from the prime factorizations.

$$\operatorname{lcm}(a,b) = p_1^{\max(a_1,b_1)} p_2^{\max(a_2,b_2)} \cdots p_n^{\max(a_n,b_n)}$$

No smaller number is divisible by *a* and *b* both.

Example: $\operatorname{lcm}(2^3 3^5 7^2, 2^4 3^3) = 2^{\max(3,4)}\, 3^{\max(5,3)}\, 7^{\max(2,0)} = 2^4\, 3^5\, 7^2$

- The greatest common divisor and the least common multiple of two integers are related by:

Theorem 5: Let a and b be positive integers. Then

$$ab = \gcd(a,b) \cdot \operatorname{lcm}(a,b)$$

(*proof is Exercise* 31)

Euclidean Algorithm

Euclid
(325 B.C.E. – 265 B.C.E.)

- The Euclidean algorithm is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that gcd(a,b) is equal to gcd(b,c) when $a > b$ and c is the remainder when a is divided by b.

Example: Find gcd(91, 287):

- $287 = 91 \cdot 3 + 14$ Divide 287 by 91
- $91 = 14 \cdot 6 + 7$ Divide 91 by 14
- $14 = 7 \cdot 2 + 0$ Divide 14 by 7

Stopping condition

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$

continued →

Correctness of Euclidean Algorithm

Lemma 1: Let $a = bq + r$, where a, b, q, and r are integers. Then $\gcd(a,b) = \gcd(b,r)$.

Proof:

- Suppose that d divides both a and b.
- Then d also divides $a - bq = r$ (by Theorem 1 of Section 4.1).
- Hence, any common divisor of a and b must also be any common divisor of b and r.
- Suppose that d divides both b and r.
- Then d also divides $bq + r = a$.
- Hence, any common divisor of a and b must also be a common divisor of b and r.
- Therefore, $\gcd(a,b) = \gcd(b,r)$. ◀

Correctness of Euclidean Algorithm

- Suppose that a and b are positive integers with $a \geq b$.
  Let $r_0 = a$ and $r_1 = b$.
  Successive applications of the division algorithm yields:

$$
\begin{array}{lll}
r_0 & = r_1 q_1 + r_2 & 0 \leq r_2 < r_1, \\
r_1 & = r_2 q_2 + r_3 & 0 \leq r_3 < r_2, \\
 & \cdot & \\
 & \cdot & \\
 & \cdot & \\
r_{n-2} & = r_{n-1} q_{n-1} + r_2 & 0 \leq r_n < r_{n-1}, \\
r_{n-1} & = r_n q_n \,. &
\end{array}
$$

- Eventually, a remainder of zero occurs in the sequence of terms: $a = r_0 > r_1 > r_2 > \cdots \geq 0$. The sequence can't contain more than a terms.
- By Lemma 1, $\gcd(a,b) = \gcd(r_0,r_1) = \cdots = \gcd(r_{n-1},r_n) = \gcd(\mathrm{r}_n, 0) = r_n$.
- Hence the greatest common divisor is the last nonzero remainder in the sequence of divisions.

Dividing Congruences by an Integer

- **Dividing both sides of a valid congruence by an integer does not always produce a valid congruence** (see Section 4.1).
- But dividing by an integer **relatively prime** to the modulus does produce a valid congruence:

Theorem 7: Let m be a positive integer and let a, b, and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c,m) = 1$, then $a \equiv b \pmod{m}$.

Proof: Since $ac \equiv bc \pmod{m}$, $m \mid ac - bc = c(a - b)$ by Lemma 2 and the fact that $\gcd(c,m) = 1$, it follows that $m \mid a - b$. Hence, $a \equiv b \pmod{m}$. ◀

Lecture 15:
Integer Representations

Section 4.2

Section Summary

- Integer Representations
  - Base b Expansions
  - Binary Expansions
  - Octal Expansions
  - Hexadecimal Expansions
- Base Conversion Algorithm
- Algorithms for Integer Operations

Representations of Integers

- In the modern world, we use *decimal,* or *base* 10, *notation* to represent integers. For example when we write 965, we mean $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$.
- We can represent numbers using any base b, where b is a positive integer greater than 1.
- The bases $b = 2$ (*binary*), $b = 8$ (*octal*) , and $b = 16$ (*hexadecimal*) are important for computing and communications
- The ancient Mayans used base 20 and the ancient Babylonians used base 60.

Base *b* Representations

- We can use positive integer b greater than 1 as a base, because of this theorem:

Theorem 1: Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where k is a nonnegative integer;

$a_0, a_1, \dots a_k$ are nonnegative integers less than b, and $a_k \neq 0$.

The a_j terms with $j = 0,\dots,k$ are called the **base-*b* digits** of the representation.

- The representation of n given in Theorem 1 is called the ***base b expansion of n*** and is denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$.

Base b Representations

From the division algorithm, we can write

$n = b \cdot q_0 + r_0$; where $0 \leq r_0 < b$ and $q_0 < n$.

Now, q_0 can also be written as: $q_0 = b \cdot q_1 + r_1$, where $q_1 < q_0$.

and consequently, $q_1 = b \cdot q_2 + r_2$, where $q_2 < q_1$, and so on.

Eventually, after k steps, q_k **will be zero**. So, $q_{k-1} = b \cdot 0 + r_k = r_k$.

Thus we have,

$$n = b \cdot q_0 + r_0 = b \cdot (b \cdot q_1 + r_1) + r_0 = b^2 \cdot q_1 + b \cdot r_1 + r_0$$

$$= b^2 \cdot (b \cdot q_2 + r_2) + b \cdot r_1 + r_0 = b^3 \cdot q_2 + b^2 \cdot r_2 + b \cdot r_1 + r_0$$

$$= b^{k+1} \cdot q_k + b^k \cdot r_k + b^{k-1} \cdot r_{k-1} + \ldots\ldots + b^3 \cdot r_3 + b^2 \cdot r_2 + b \cdot r_1 + r_0$$

$$= b^k \cdot r_k + b^{k-1} \cdot r_{k-1} + \ldots\ldots + b^3 \cdot r_3 + b^2 \cdot r_2 + b \cdot r_1 + r_0 \; [q_k = 0]$$

Binary Expansions

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

Example: What is the decimal expansion of the integer that has

$(1\ 0101\ 1111)_2$ as its binary expansion?

Solution: $(1\ 0101\ 1111)_2$

$= 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 351.$

Example: What is the decimal expansion of the integer that has $(11011)_2$ as its binary expansion?

Solution: $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27.$

Octal Expansions

The octal expansion (base 8) uses the digits {0,1,2,3,4,5,6,7}.

Example: What is the decimal expansion of the number with octal expansion $(7016)_8$?

Solution: $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

Example: What is the decimal expansion of the number with octal expansion $(111)_8$?

Solution: $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

Hexadecimal Expansions

The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}. The letters A through F represent the decimal numbers 10 through 15.

Example: What is the decimal expansion of the number with hexadecimal expansion $(2AE0B)_{16}$?

Solution:

$2\cdot16^4 + 10\cdot16^3 + 14\cdot16^2 + 0\cdot16^1 + 11\cdot16^0 = 175627$

Example: What is the decimal expansion of the number with hexadecimal expansion $(E5)_{16}$?

Solution: $14\cdot16^1 + 5\cdot16^0 = 224 + 5 = 229$

Base Conversion

Example: Find the octal expansion of $(12345)_{10}$

Solution: Successively dividing by 8 gives:

$$
\begin{aligned}
12345 &= 8 \cdot 1543 + 1 \\
1543 &= 8 \cdot 192 + 7 \\
192 &= 8 \cdot 24 + 0 \\
24 &= 8 \cdot 3 + 0 \\
3 &= 8 \cdot 0 + 3
\end{aligned}
$$

The remainders are the digits from right to left, yielding $(30071)_8$.

Comparison of Hexadecimal, Octal, and Binary Representations

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Initial 0s are not shown

- Each octal digit corresponds to a block of 3 binary digits.
- Each hexadecimal digit corresponds to a block of 4 binary digits.
  So, conversion between binary, octal, and hexadecimal is easy.

Conversion Between Binary, Octal, and Hexadecimal Expansions

Example: Find the octal and hexadecimal expansions of $(11\ 1110\ 1011\ 1100)_2$.

Solution:

- To convert to octal, we group the digits into blocks of three, adding initial 0s as needed:

  $(011\quad 111\quad 010\quad 111\quad 100)_2$

  The blocks from left to right correspond to the digits 3, 7, 2, 7 and 4. Hence, the solution is $(37274)_8$.

- To convert to hexadecimal, we group the digits into blocks of four, adding initial 0s as needed:

  $(0011\quad 1110\quad 1011\quad 1100)_2$

  The blocks from left to right correspond to the digits 3, E, B and C. Hence, the solution is $(3EBC)_{16}$.

Binary (Fast) Modular Exponentiation

In cryptography, it is important to be able to find $b^n \mathbf{mod}\ m$ efficiently, where b, n, and m are large integers.

- We can compute the exponential using the fact:

$$b^{s+t} \mathbf{mod}\ m = (b^s \mathbf{mod}\ m)(b^t \mathbf{mod}\ m)\ \mathbf{mod}\ m$$

- Also, binary representation of n yields:

$$n = a_{k-1}2^{k-1} + \ldots + a_1 2 + a_0 = (a_{k-1} \ldots a_1\, a_0)_2$$

  - Note that Here $a_k, a_{k-1}, \ldots, a_1, a_0 \in \{0, 1\}$.

- Then, $b^n = b^{a_{k-1}\cdot 2^{k-1}+\cdots+a_1\cdot 2+a_0} = b^{a_{k-1}\cdot 2^{k-1}} \cdots b^{a_1\cdot 2} \cdot b^{a_0}$

- To calculate $b^n \mathbf{mod}\ m$, successively find:
  $b\ \mathbf{mod}\ m$, $b^2\ \mathbf{mod}\ m$, $b^4\ \mathbf{mod}\ m$, $\ldots$, $b^{2^\wedge(k-1)}\ \mathbf{mod}\ m$
  Then, multiply together the terms $b^{2^\wedge j} \mathbf{mod}\ m$ where $a_j = 1$.

Example: Compute 3^{11} using this method.
Solution: Note that $11 = (1011)_2$ so that $3^{11} = 3^8\, 3^2\, 3^1 = ((3^2)^2)^2\, 3^2\, 3^1$
$= (9^2)^2 \cdot 9 \cdot 3 = (81)^2 \cdot 9 \cdot 3 = 6561 \cdot 9 \cdot 3 = 117{,}147.$

Fast Modular Exponentiation Example

Example: Find 3^{144} **mod** 100.

Solution: Here, $144 = (10010000)_2$ so $3^{144} = 3^{128}\,3^{16}$. Now,

- $3 \textbf{ mod } 100 = 3$
- $3^2 \textbf{ mod } 100 = (3 * 3) \textbf{ mod } 100 = 9$
- $3^4 \textbf{ mod } 100 = (9 * 9) \textbf{ mod } 100 = 81$
- $3^8 \textbf{ mod } 100 = (81 * 81) \textbf{ mod } 100$
  $= 6561 \textbf{ mod } 100 = 61$
- $3^{16} \textbf{ mod } 100 = (61 * 61) \textbf{ mod } 100$
  $= 3721 \textbf{ mod } 100 = 21$
- $3^{32} \textbf{ mod } 100 = (21 * 21) \textbf{ mod } 100$
  $= 441 \textbf{ mod } 100 = 41$
- $3^{64} \textbf{ mod } 100 = (41 * 41) \textbf{ mod } 100$
  $= 1681 \textbf{ mod } 100 = 81$
- $3^{128} \textbf{ mod } 100 = (81 * 81) \textbf{ mod } 100$
  $= 6561 \textbf{ mod } 100 = 61$

So, $3^{144} \textbf{ mod } 100 = (3^{128} \textbf{ mod } 100)(3^{16} \textbf{ mod } 100) \textbf{ mod } 100$

$= (61*21) \textbf{ mod } 100 = 1281 \textbf{ mod } 100 = 81.$

Therefore, $3^{144} \textbf{ mod } 10 = 81$ (Answer)